

07. Visualization Design Principles (Heuristics)

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<http://info-design-lab.github.io/>

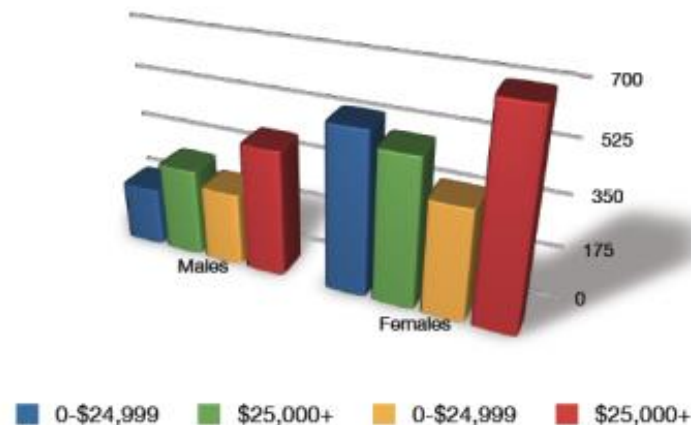
Tufte's design principles for graphical excellence

1. Maximize the data-ink ratio, within reason
2. Mobilize every graphical element, perhaps several times over, to show the data
3. Maximize data density and the size of the data matrix, within reason
4. Establish context
5. Show cause and effect, where possible
6. Compare and contrast, utilize layering & separation
7. Escape flatland, use small multiples, parallel sequencing (reality is multivariate)
8. Show multiple dimensions
9. Utilize narratives of space and time
10. Integrate image, number and text

1. Maximize the data-ink ratio, within reason

- Data-ink is the non-erasable core of a graphic, the non-redundant ink arranged in response to variation in the numbers represented
- It is also the proportion of graphic's ink devoted to the non-redundant display of data-information.

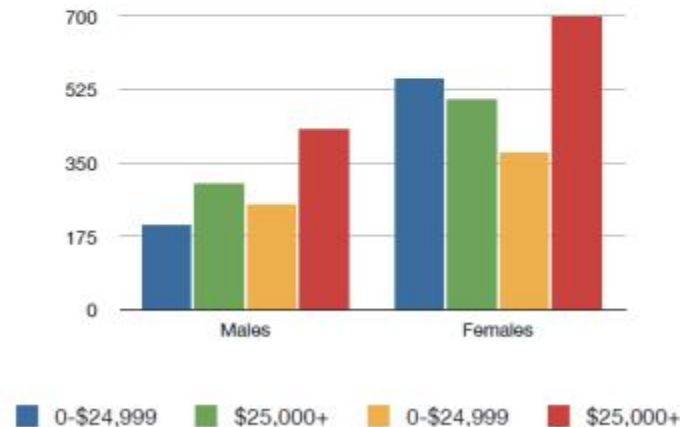
$$\text{Data-Ink Ratio} = \frac{\text{Data ink}}{\text{Total ink used in graphic}}$$



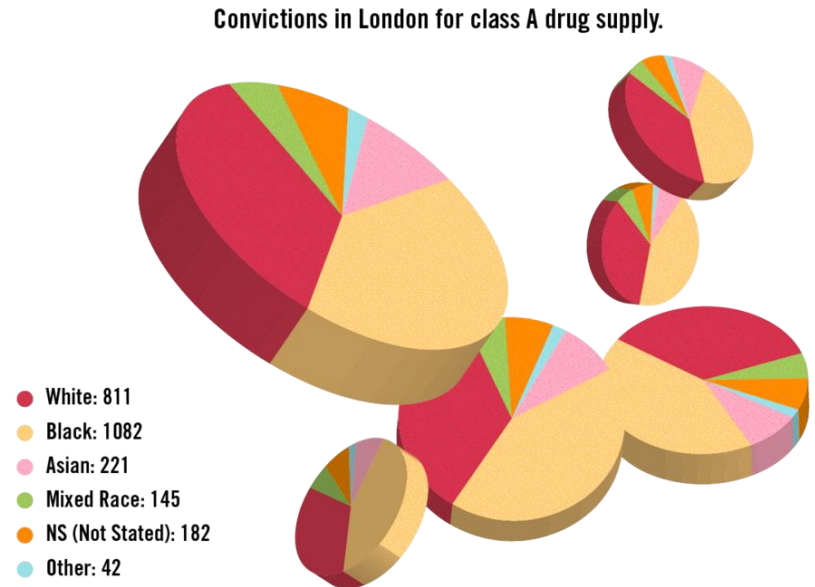
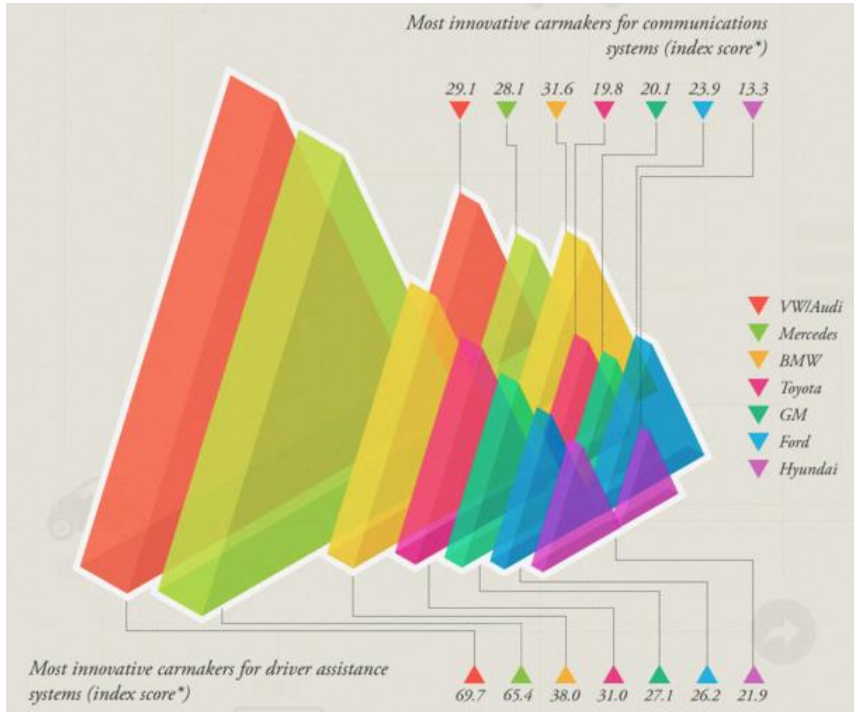
1. Maximize the data-ink ratio, within reason

- This rule states that a visualization should contain as much data as possible while also using as little pixels as possible
- Through a comprehensive editing and testing process, any visualization can continually be improved upon

$$\text{Data-Ink Ratio} = \frac{\text{Data ink}}{\text{Total ink used in graphic}}$$



Unjustified 3D all too common, in the news and elsewhere



<http://viz.wtf/post/137826497077/eye-popping-3d-triangles>

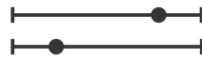
<http://viz.wtf/post/139002022202/designer-drugs-ht-ducqn>

Depth vs power of the plane

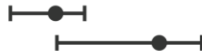
- high-ranked spatial position channels: **planar** spatial position
– not depth!

➔ Magnitude Channels: Ordered Attributes

Position on common scale



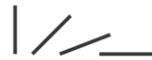
Position on unaligned scale



Length (1D size)



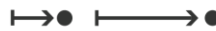
Tilt/angle



Area (2D size)



Depth (3D position)



Depth vs power of the plane

- high-ranked spatial position channels: **planar** spatial position
– not depth!

➔ Magnitude Channels: Ordered Attributes

Position on common scale



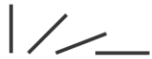
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Length (1D size)



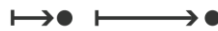
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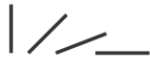
Position on unaligned scale



Length (1D size)



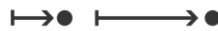
Tilt/angle



Area (2D size)



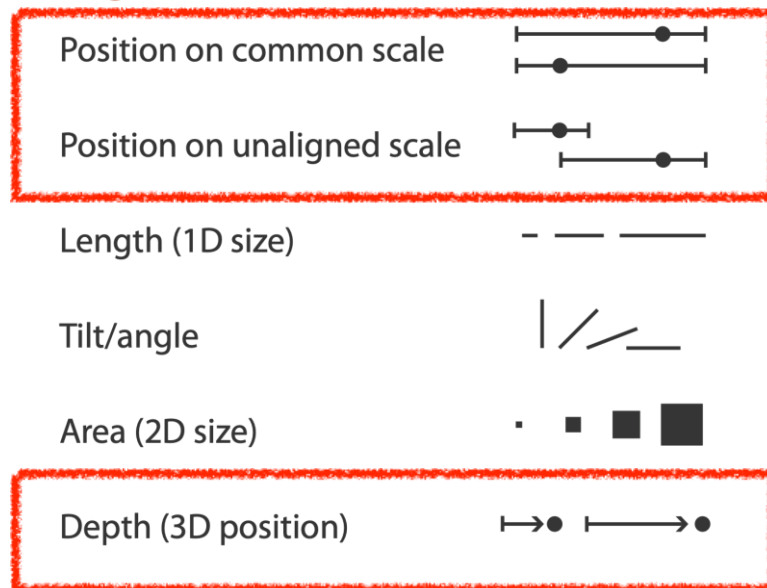
Depth (3D position)



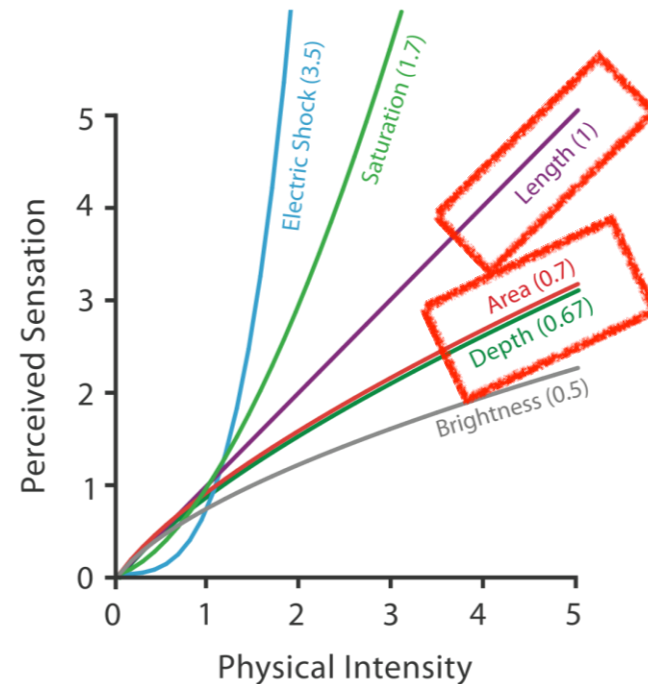
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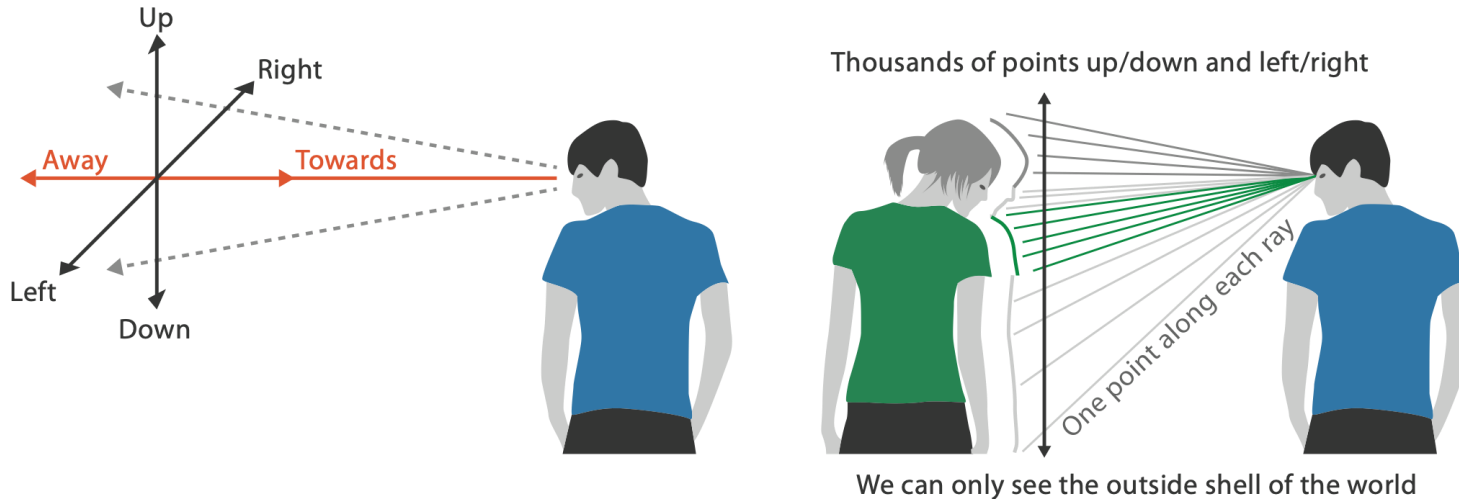


Steven's Psychophysical Power Law: $S = I^N$



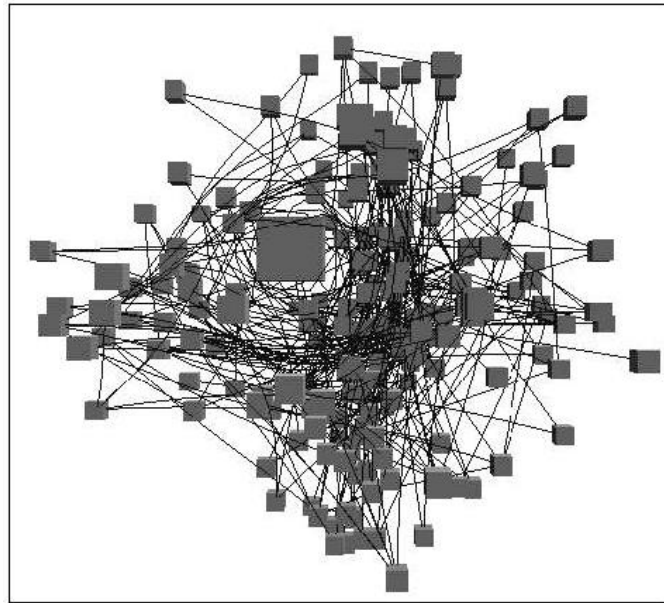
No unjustified 3D: Danger of depth

- we don't really live in 3D: we **see** in 2.05D
 - acquire more info on image plane quickly from eye movements
 - acquire more info for depth slower, from head/body motion



Occlusion hides information

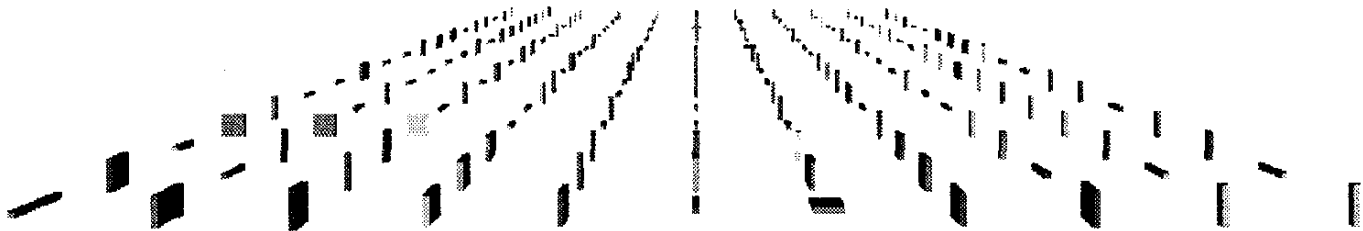
- occlusion
- interaction can resolve, but at cost of time and cognitive load



[Distortion Viewing Techniques for 3D Data. Carpendale et al. InfoVis1996.]

Perspective distortion loses information

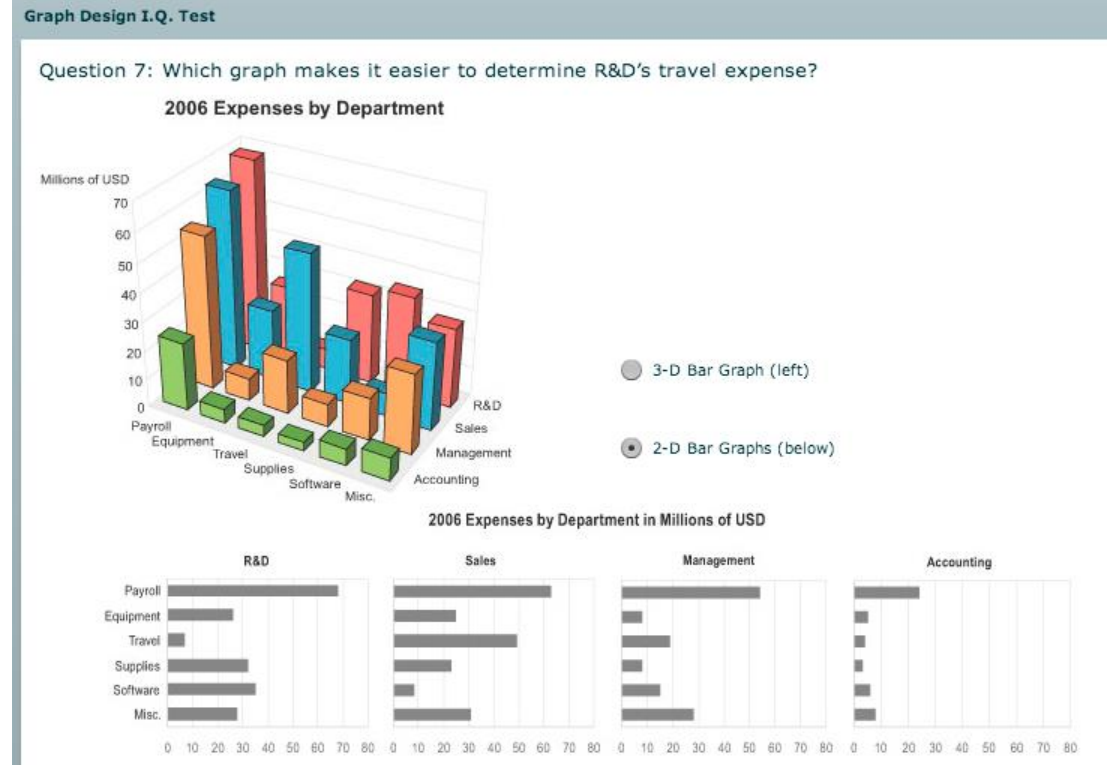
- perspective distortion
 - interferes with all size channel encodings
 - power of the plane is lost!



[Visualizing the Results of Multimedia Web Search Engines. Mukherjea, Hirata, and Hara. InfoVis 96]

3D vs 2D bar charts

- 3D bars:
 - very difficult to justify!
 - perspective distortion
 - occlusion
- faceting into 2D almost always better choice



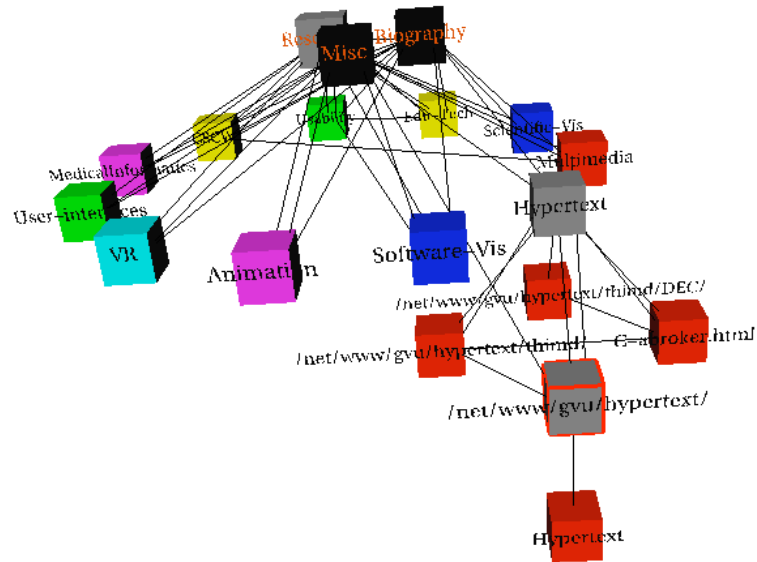
<http://perceptualedge.com/files/GraphDesignIQ.html>

Tilted text isn't legible

- text legibility
 - far worse when tilted from image plane

- further reading

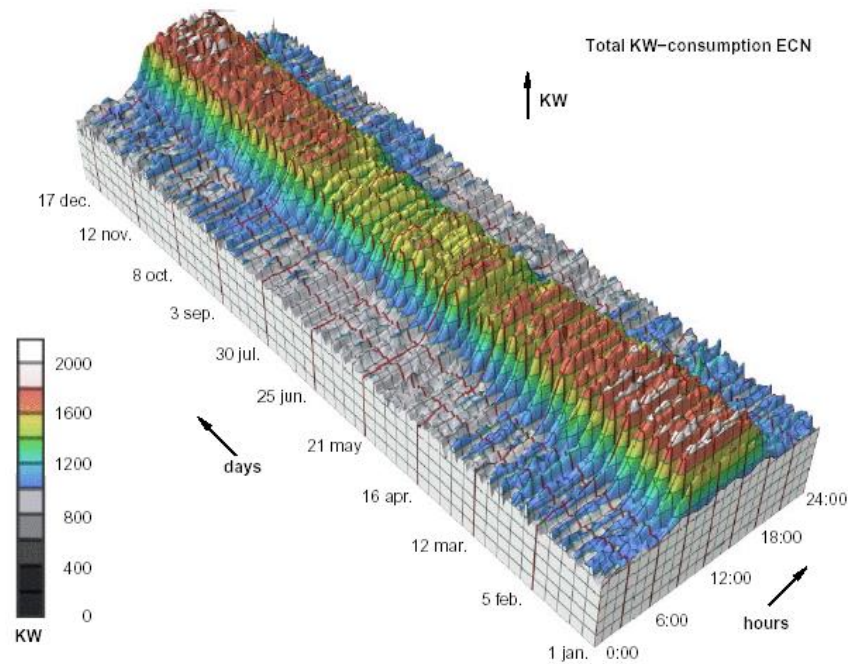
Exploring and Reducing the Effects of Orientation on Text Readability in Volumetric Displays.
Grossman et al. CHI 2007



[Visualizing the World-Wide Web with the Navigational View Builder.
Mukherjea and Foley. Computer Networks and ISDN Systems, 1995.]

No unjustified 3D example: Time-series data

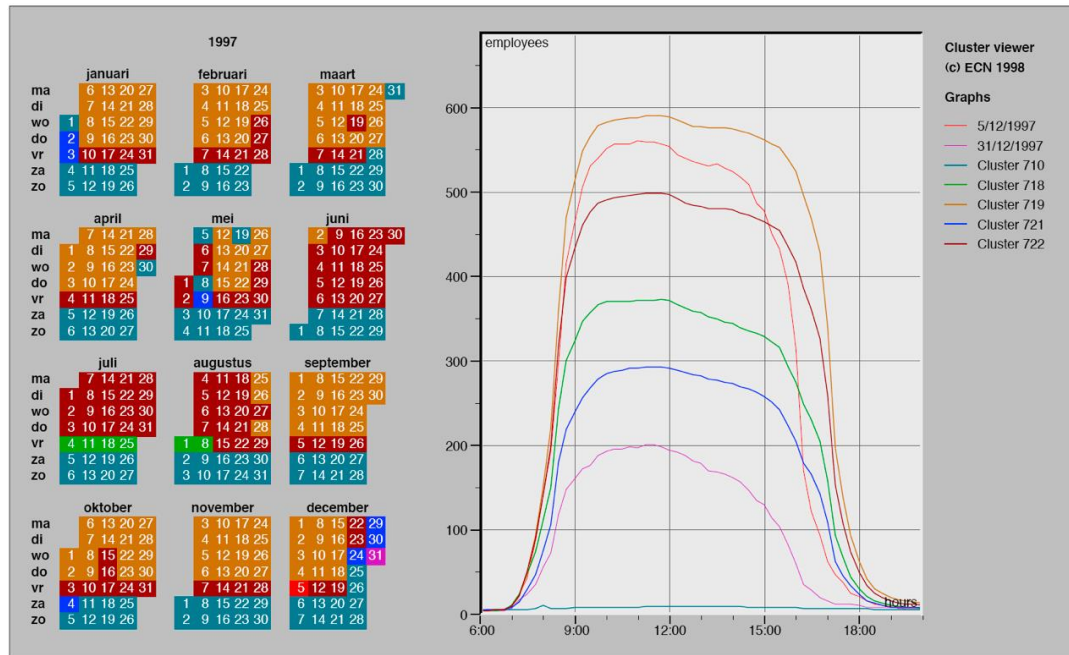
- extruded curves: detailed comparisons impossible



[Cluster and Calendar based Visualization of Time Series Data. van Wijk and van Selow, Proc. InfoVis 99.]

No unjustified 3D example: Transform for new data abstraction

- derived data: cluster hierarchy
- juxtapose multiple views: calendar, superimposed 2D curves



[Cluster and Calendar based Visualization of Time Series Data. van Wijk and van Selow, Proc. InfoVis 99.]

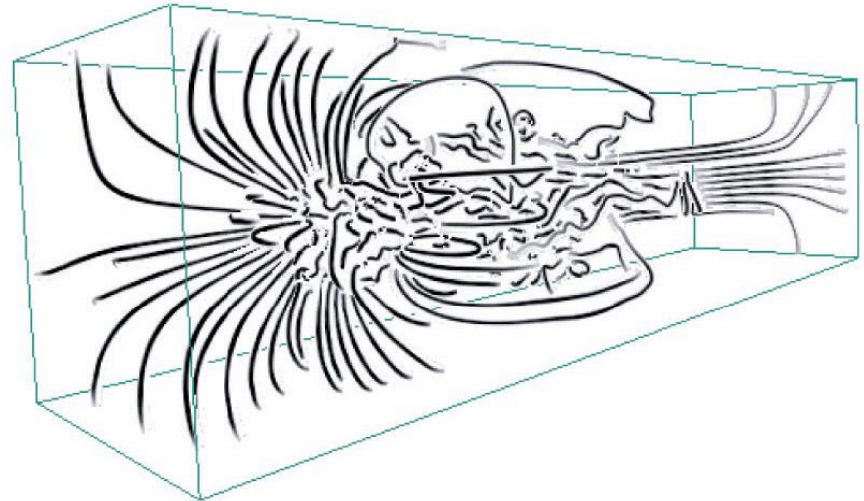
Justified 3D: shape perception

- benefits outweigh costs when task is shape perception for 3D spatial data
 - interactive navigation supports synthesis across many viewpoints

Targets

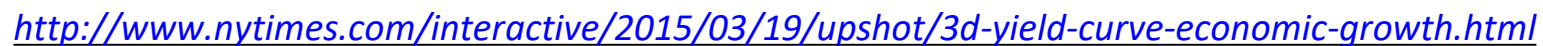
➞ Spatial Data

→ Shape

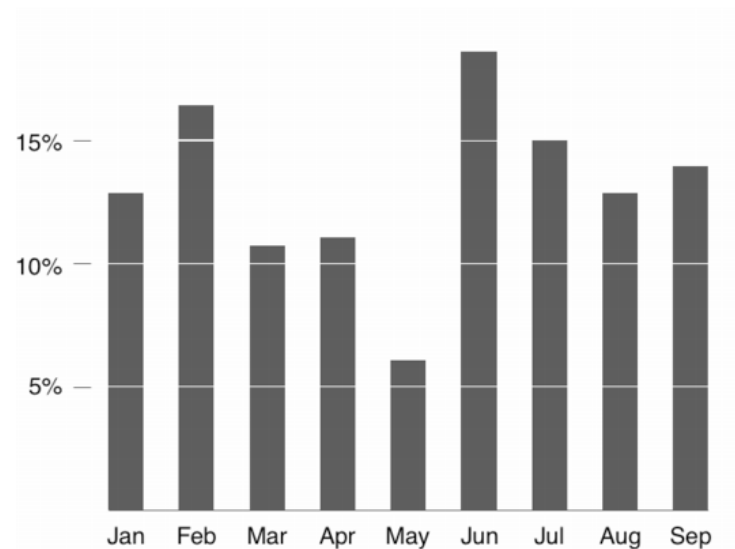
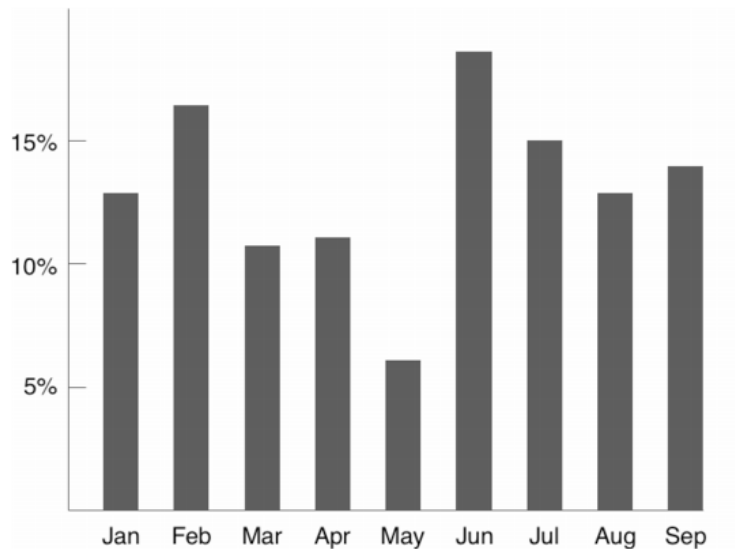
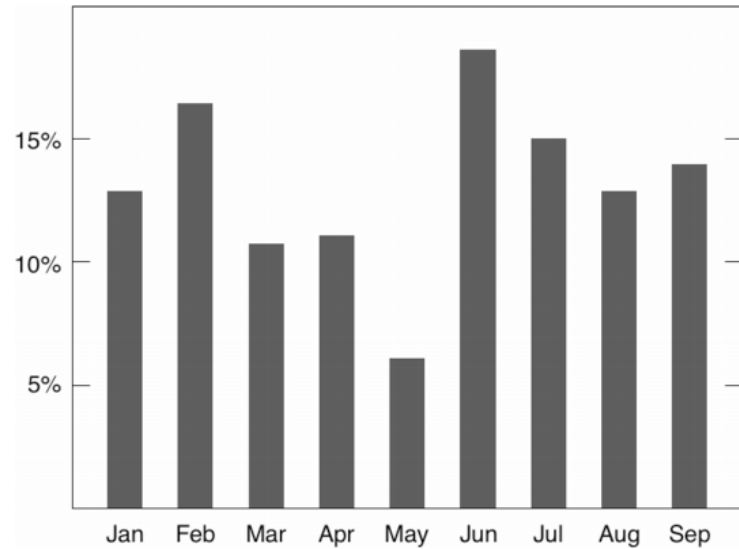
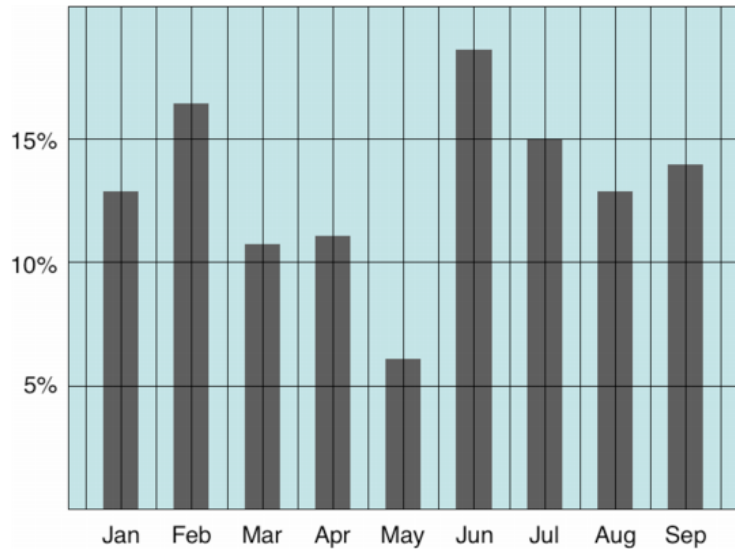


*[Image-Based Streamline Generation and Rendering. Li and Shen.
IEEE Trans. Visualization and Computer Graphics (TVCG) 13:3 (2007), 630–640.]*

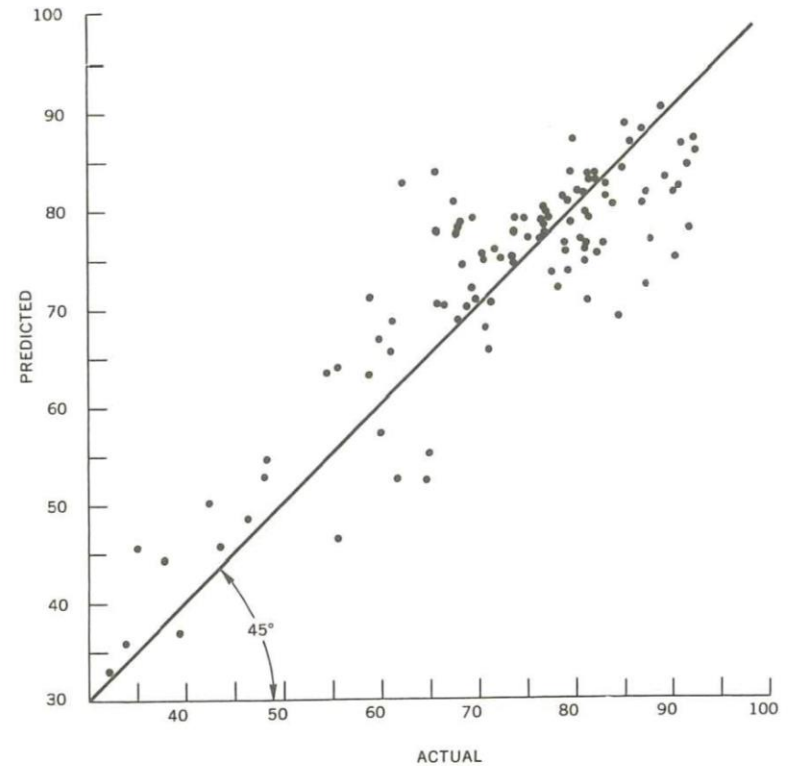
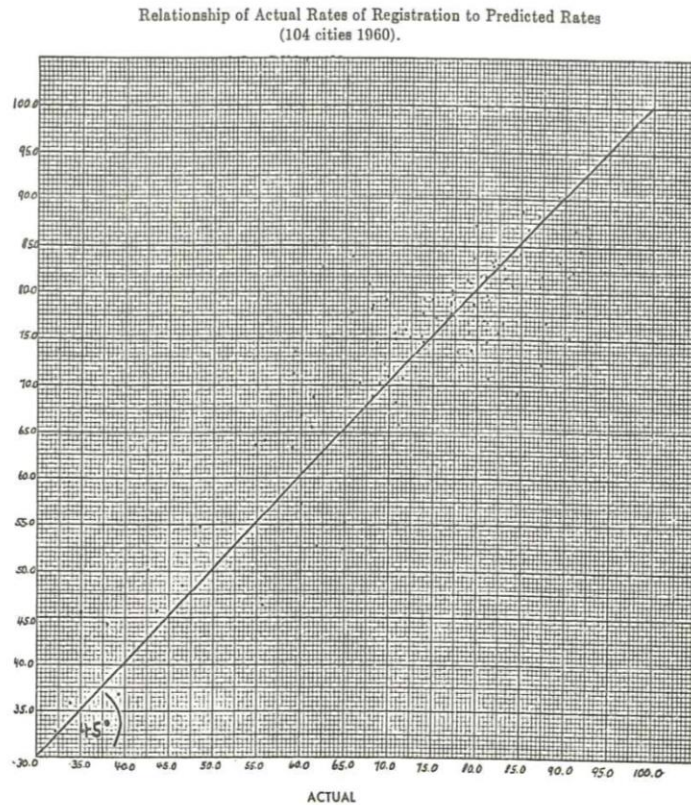
- constrained navigation steps through carefully designed viewpoints



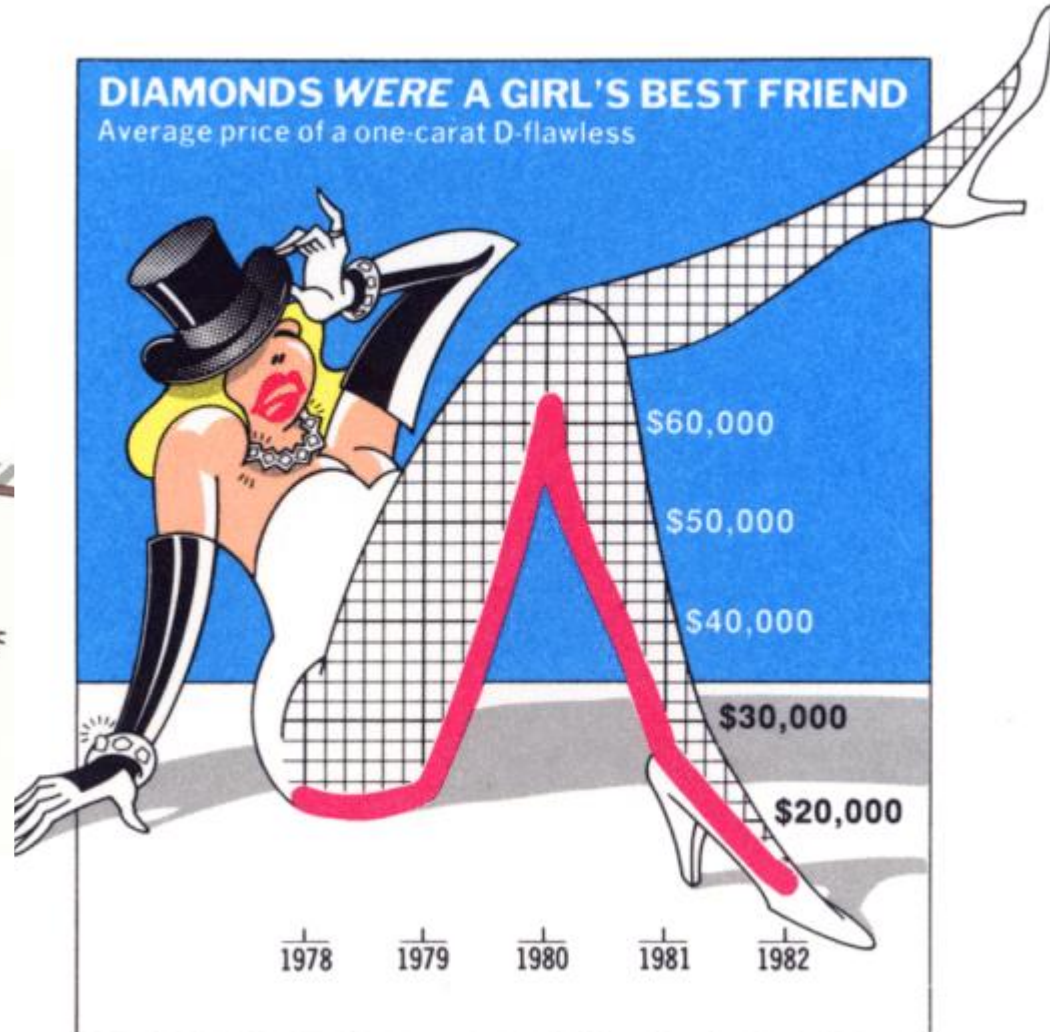
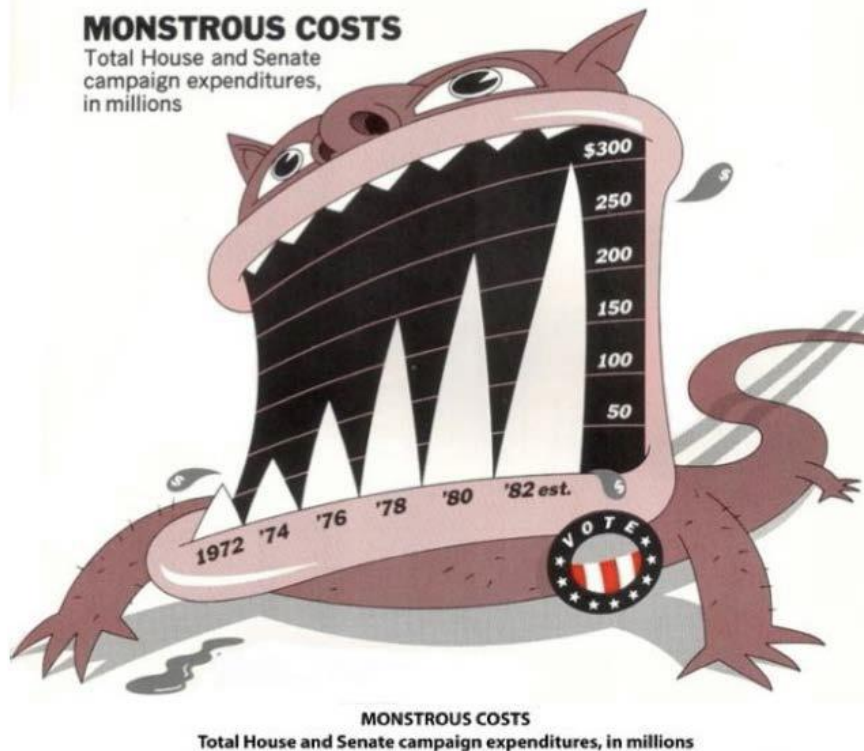
1. Maximize the data-ink ratio, within reason



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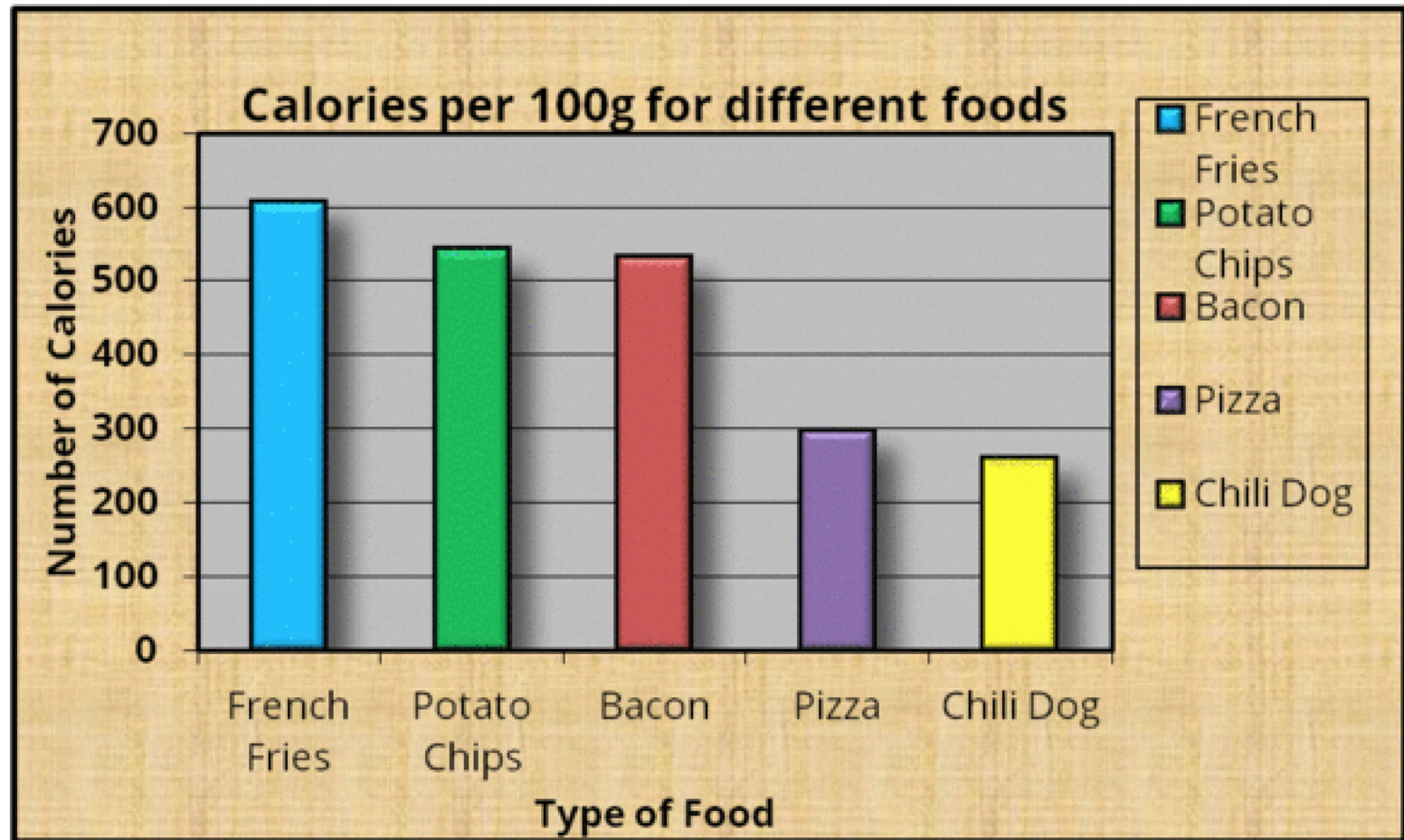


1. Maximize the data-ink ratio, within reason

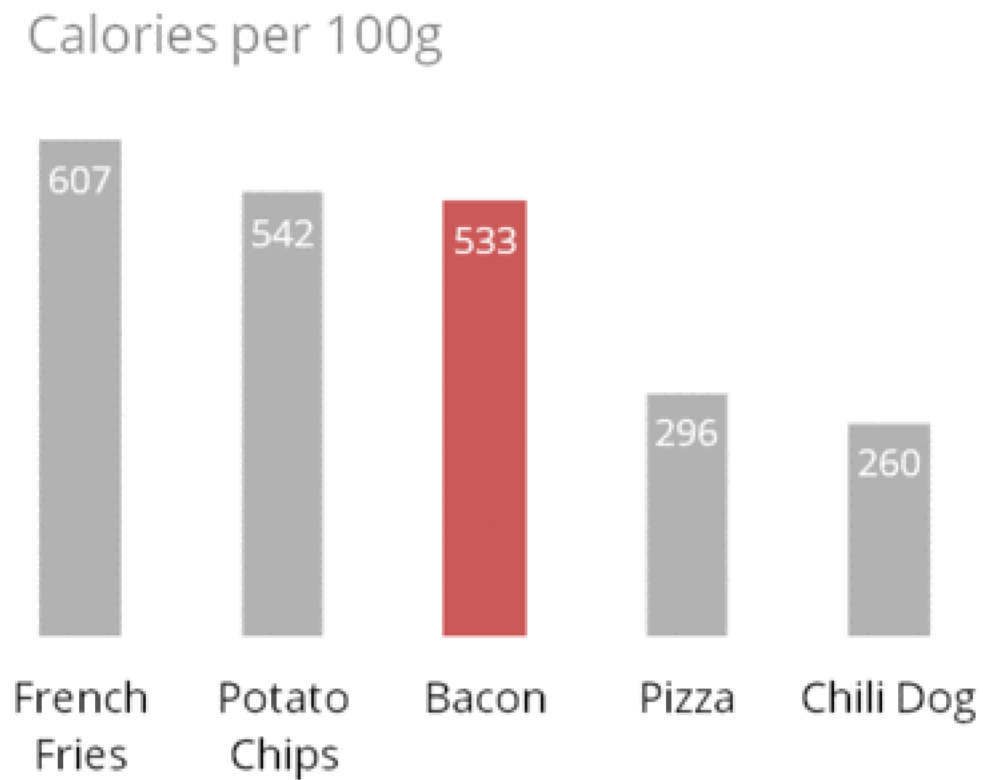


Nigel Holmes

1. Maximize the data-ink ratio, within reason

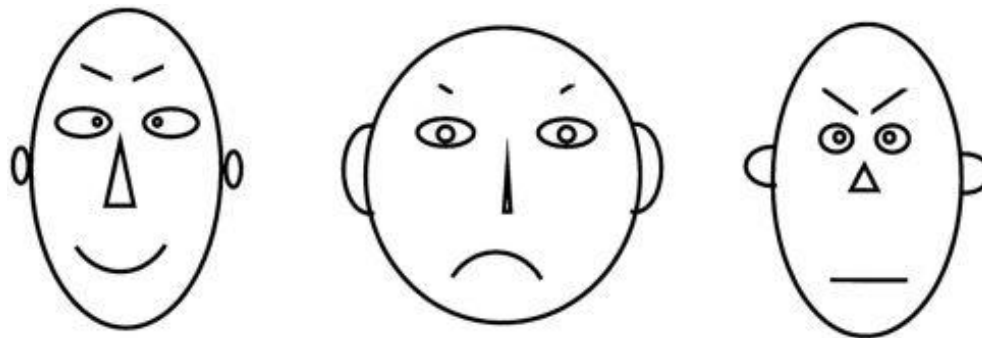


1. Maximize the data-ink ratio, within reason



2. Mobilize every graphical element to show the data

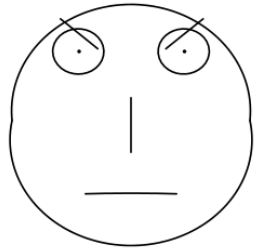
- The danger of multifunctioning elements is that they tend to generate graphical puzzles, with encodings that can only be broken by their designer.
- Thus design techniques for enhancing graphical clarity in the face of complexity must be developed along with multifunctioning elements.
- In other words, we should try to make all present graphical elements data encoding elements. We must make every graphical element effective.



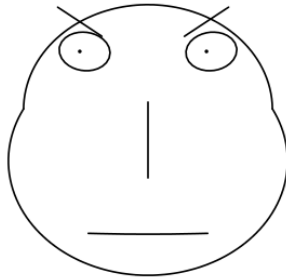
2. Mobilize every graphical element to show the data

- Invented by applied mathematician, statistician and physicist Herman Chernoff in 1973, display multivariate data in the shape of a human face.
- The individual parts, such as eyes, ears, mouth and nose represent values of the variables by their shape, size, placement and orientation.
- The key idea is that humans are well trained to recognize faces and discern small changes without difficulty. They allow for easy outlier detection and pattern recognition despite multiple dimensions of the data.
- Faces are also emotionally expressive, and thus could be used to semantically resonant encode data.

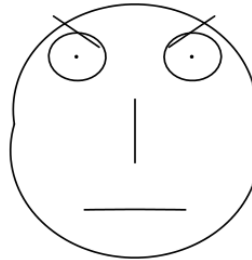
2. Mobilize every graphical element to show the data



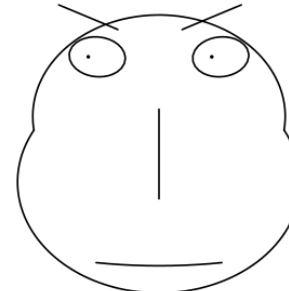
AARONSON, L.H.



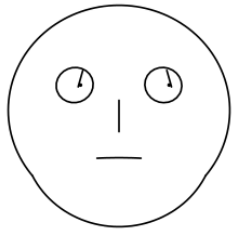
ALEXANDER, J.M.



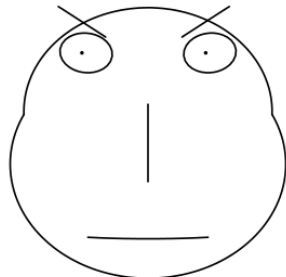
ARMENTANO, A.J.



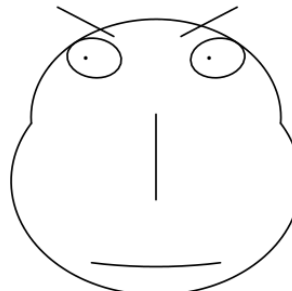
BERDON, R.I.



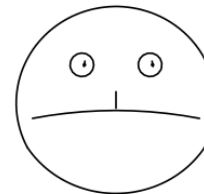
BRACKEN, J.J.



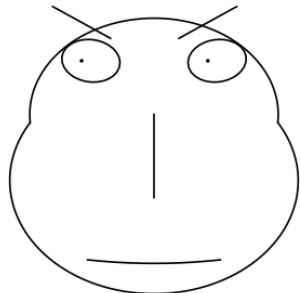
BURNS, E.B.



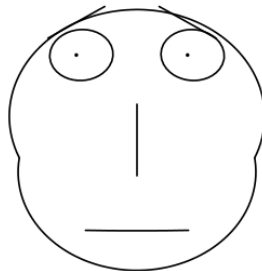
CALLAHAN, R.J.



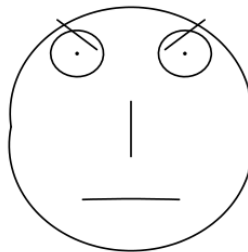
COHEN, S.S.



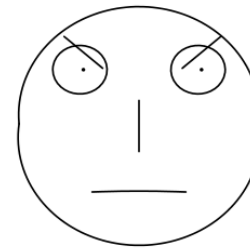
DALY, J.J.



DANNEHY, J.F.



DEAN, H.H.

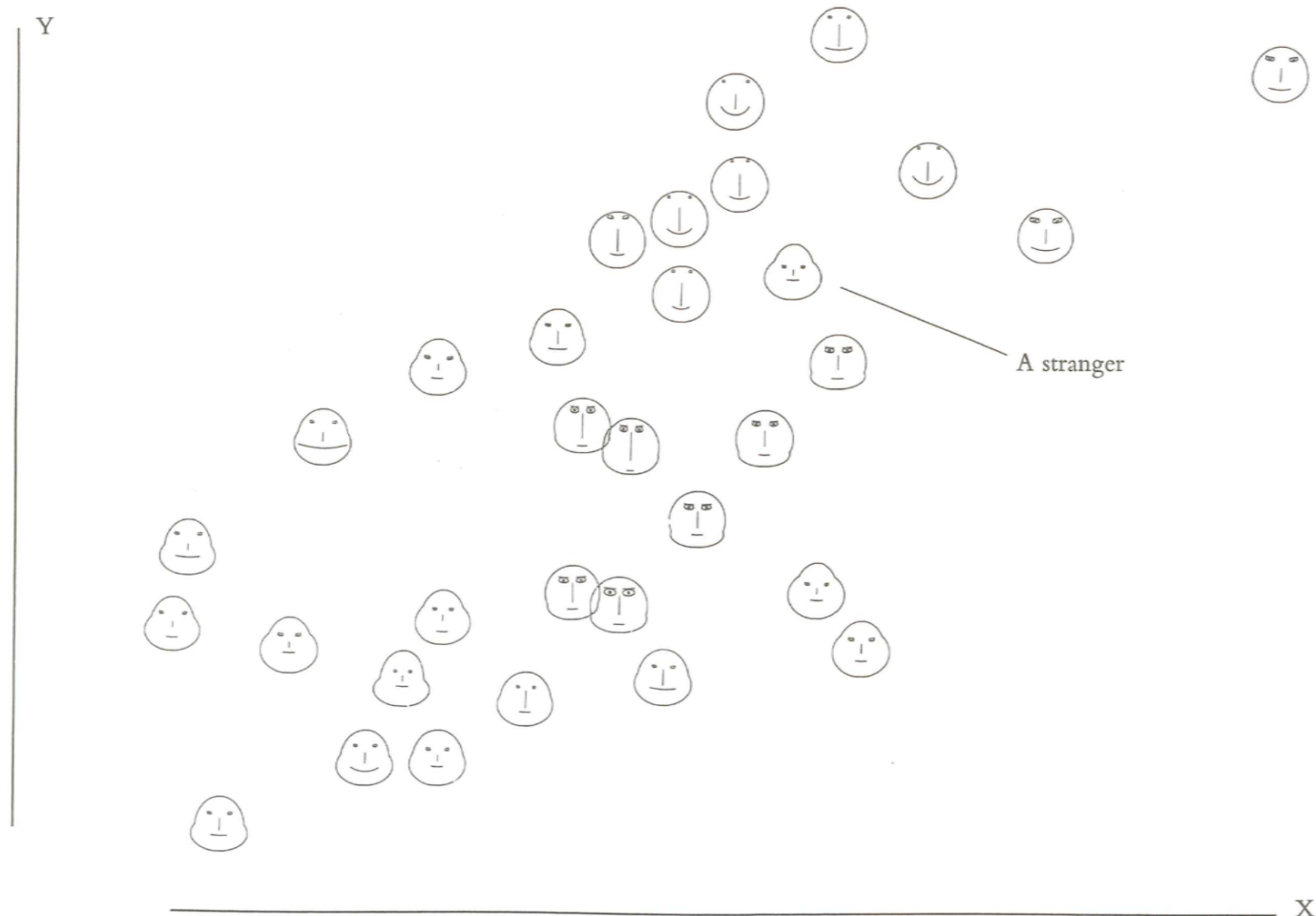


DEVITA, H.J.

Chart showing Chernoff faces for data selected from a dataset, which contains ratings of state judges in the US Superior Court by lawyers who have had contact with them.

2. Mobilize every graphical element to show the data

Chernoff Faces

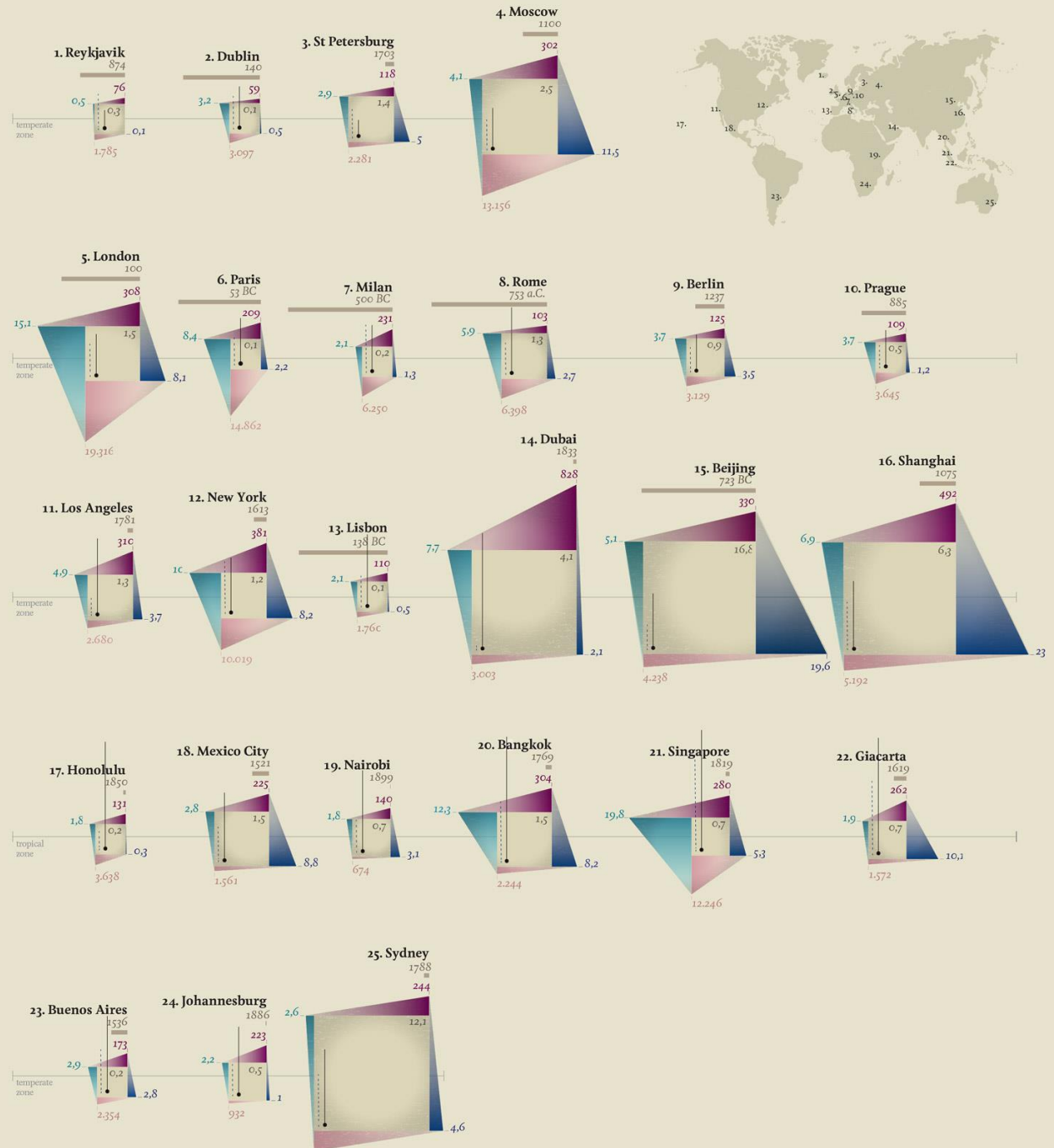
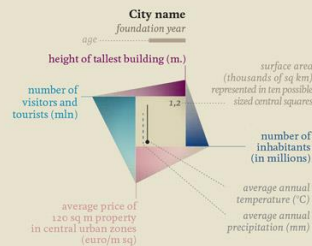


Urban story: Lisbon is on a par with Honolulu

Among 25 of the world's most important cities are visualised here. Each metropolis has been grouped vertically by 5 main temperature zones according to their latitude; and ordered horizontally according to their longitude. Each city identity is visually represented in a multiform polygon correlating to its size in sq m, number of inhabitants, number of tourists per year, height of the tallest building, and the average price of property according to real estate parameters (square metres). Further information is provided for each city "id-card": age of the city, average temperature and rainfall. We discover that latitude and longitude parameters not only determine environmental factors but also several urban characteristics. Surprisingly Lisbon and Honolulu have a lot in common!

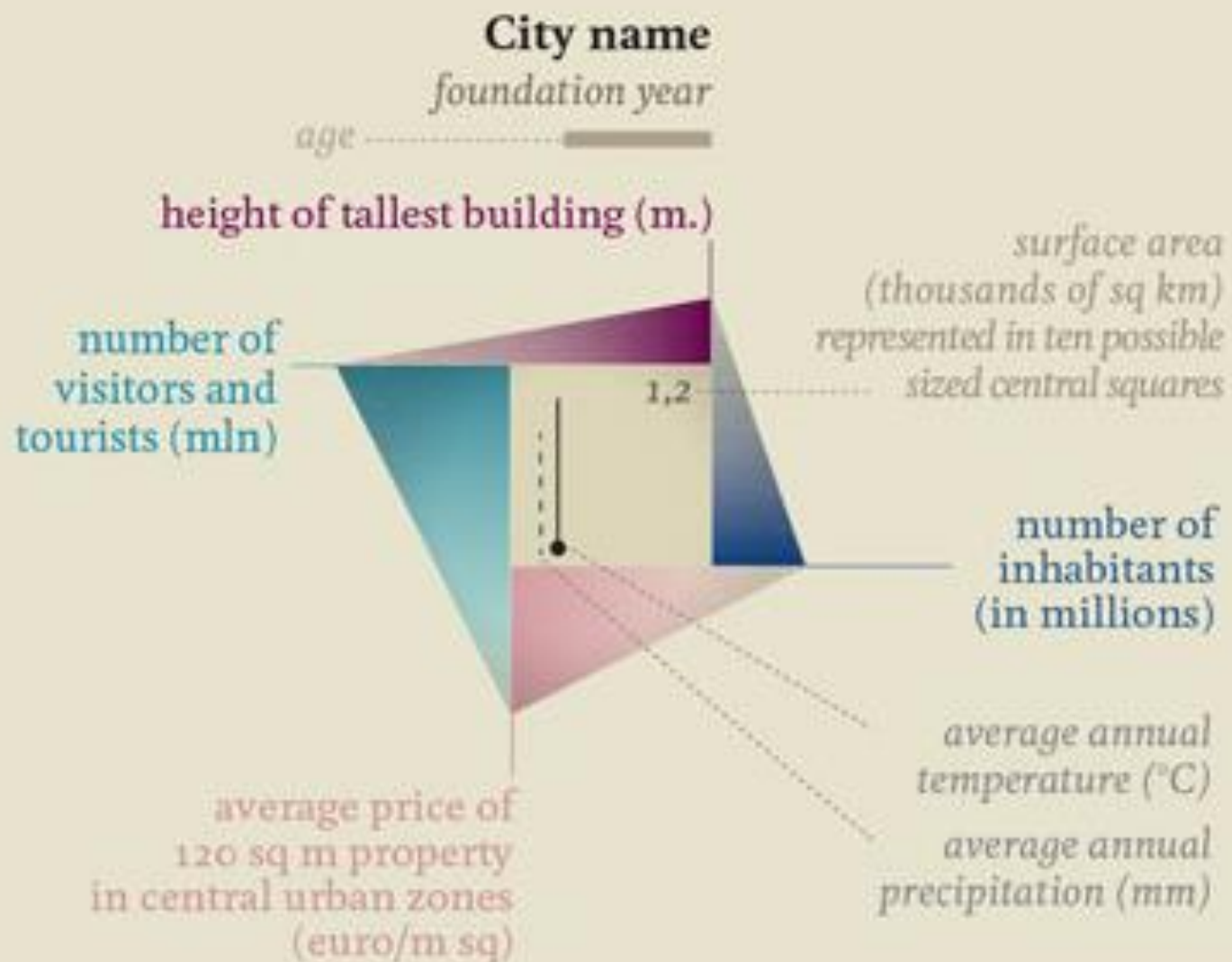
Sources:
city-data.com, currentresults.com,
euromonitor.com, globalpropertyguide.com,
skyscrapercenter.com, weatherbase.com,
wikipedia.org.

How to read it?



The visualization has been designed and produced by Accurat (www.accurat.it), and was originally published in Italian on La Lettura the Sunday cultural supplement of Corriere della Sera.

How to read it?



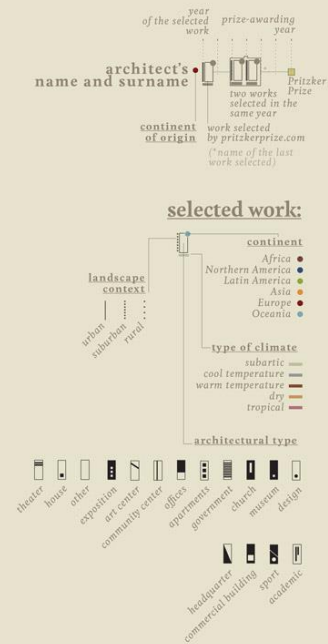
The Pritzker Architecture Prize

The visualization explores the Pritzker Prize assignment evolution, since 1979 to 2013. For each winner we visualized biographical information such as the continent of origin, the year of birth and death, the prize-awarding year and the age at the moment of the awards. Visualized are also the body of work selected by pritzkerprize.com per each architect, representing the architectural type, the year, the landscape context, the continent and the type of climate.

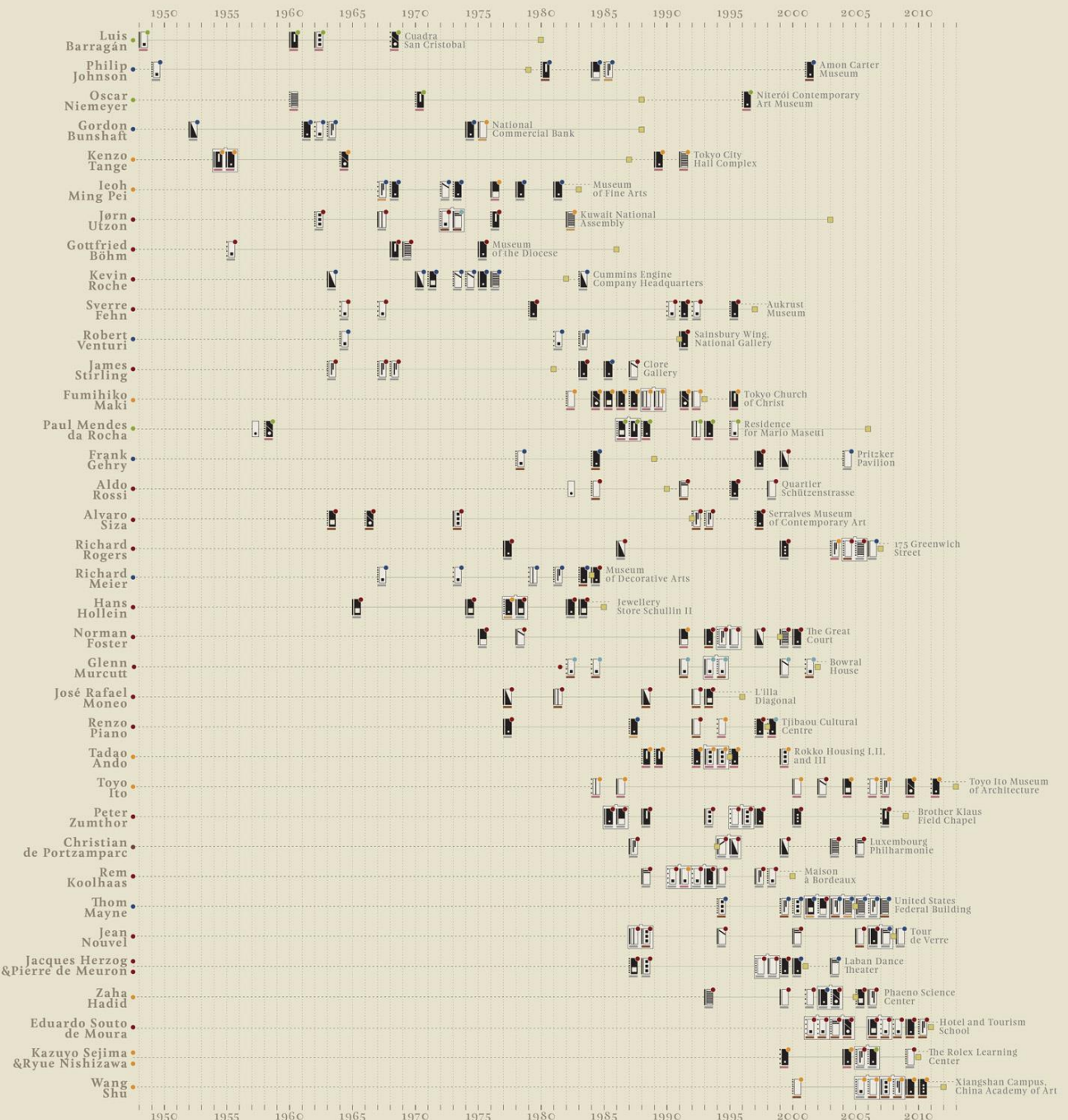
Sources: archiplanet.org, ees.rochester.edu, greatbuildings.com, pritzkerprize.com

How to read it?

Architects are vertically ordered by age, from the oldest to the youngest

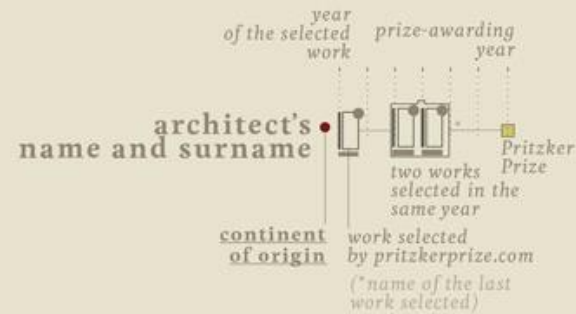


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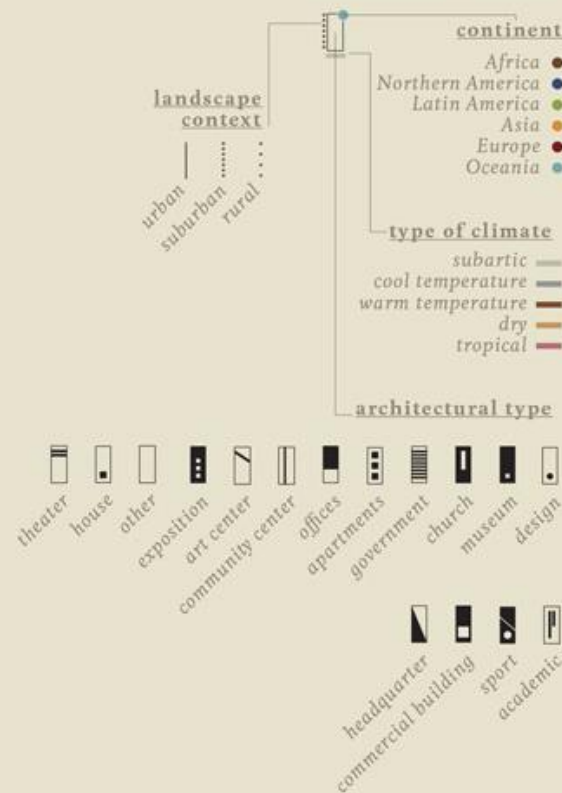


How to read it?

Architects are vertically ordered by age,
from the oldest to the youngest



selected work:



From first published to masterpieces

Visualized are the authors of the 100 best english novels of 1900 according to the Modern Library ranking.

Authors are represented through circles showing their life span where their debut novel, their masterpiece(s) according to the ranking are positioned.

Authors' order points out the time-gap between the debut novel and the first published novel selected from the Modern Library ranking (masterpiece).

Debut novel and first masterpiece correspond for the first 22 authors. Information about authors' hometown and novels' ranking are reported as well.

Sources:

biography.com, britannica.com, modernlibrary.com

How to read it?

Authors are ordered from the earliest success to the last one.



The visualization has been designed and produced by Accurat (www.accurat.it), and was originally published in Italian on La Lettura the sunday cultural supplement of Corriere della Sera.

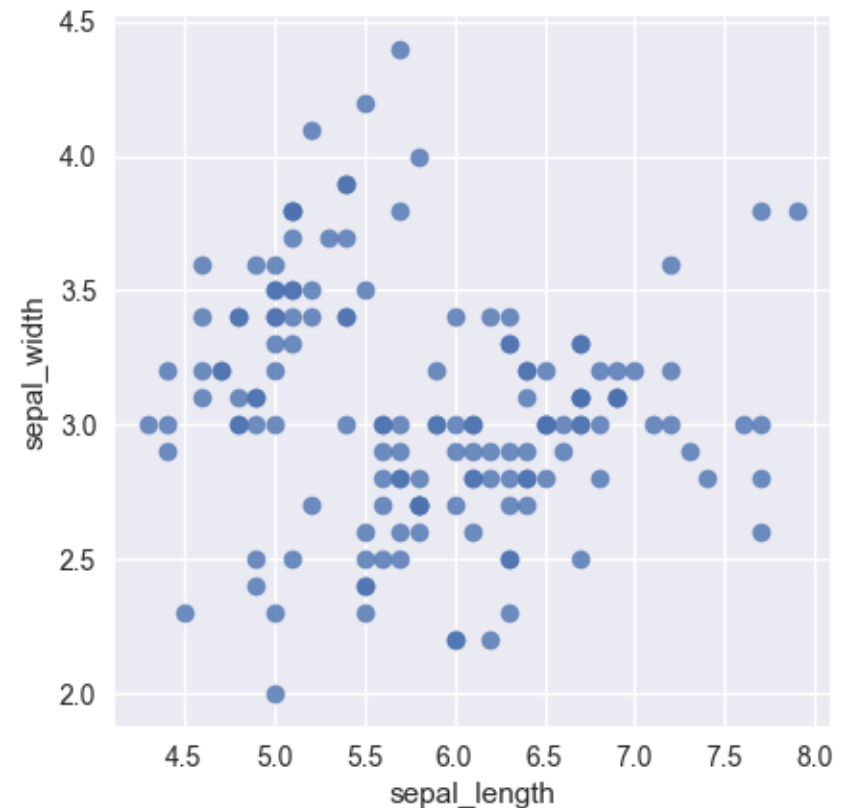
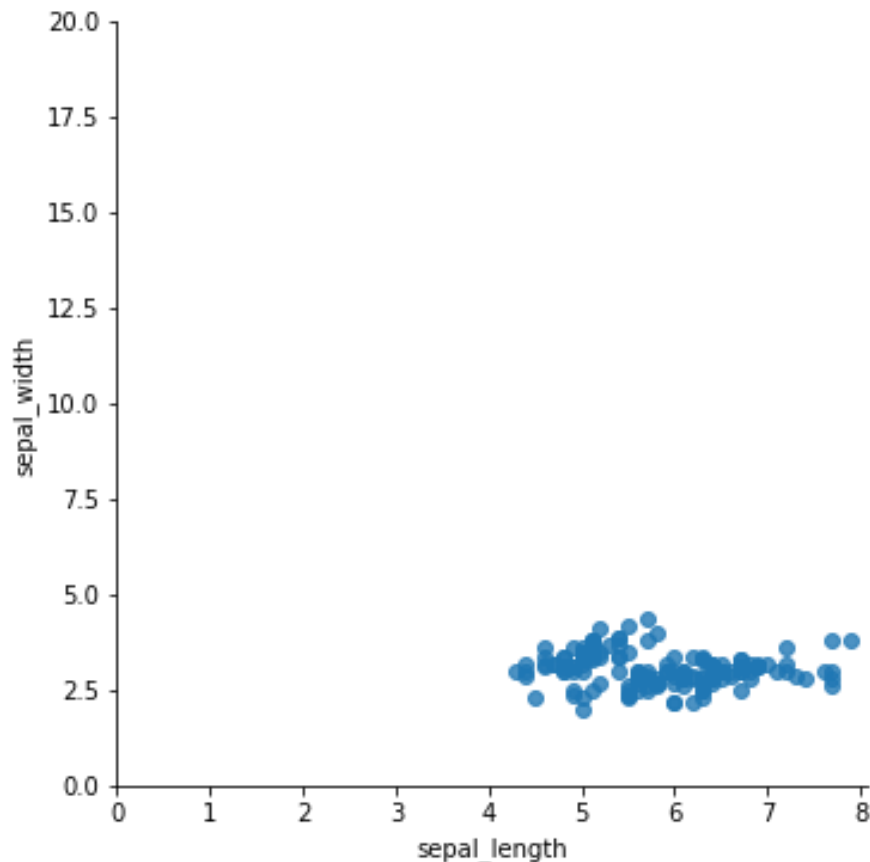
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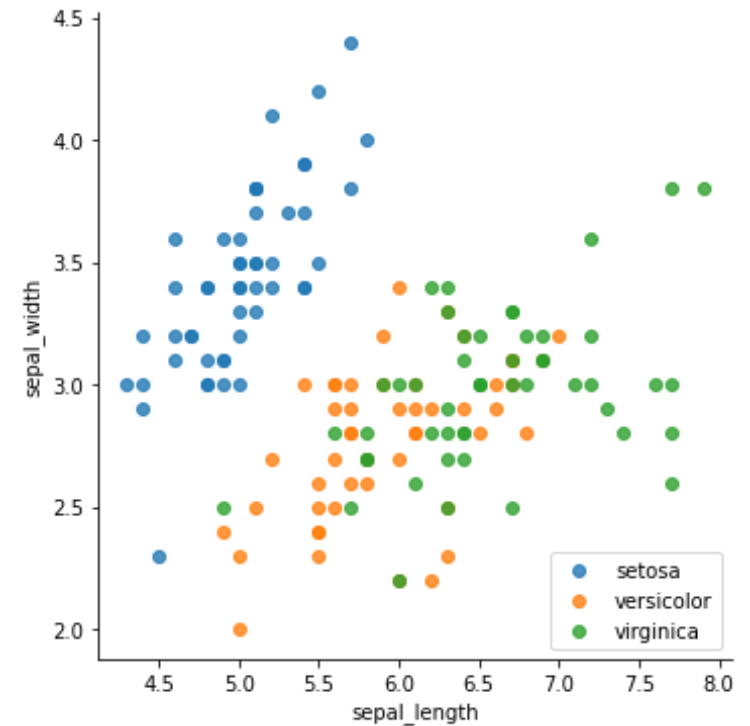
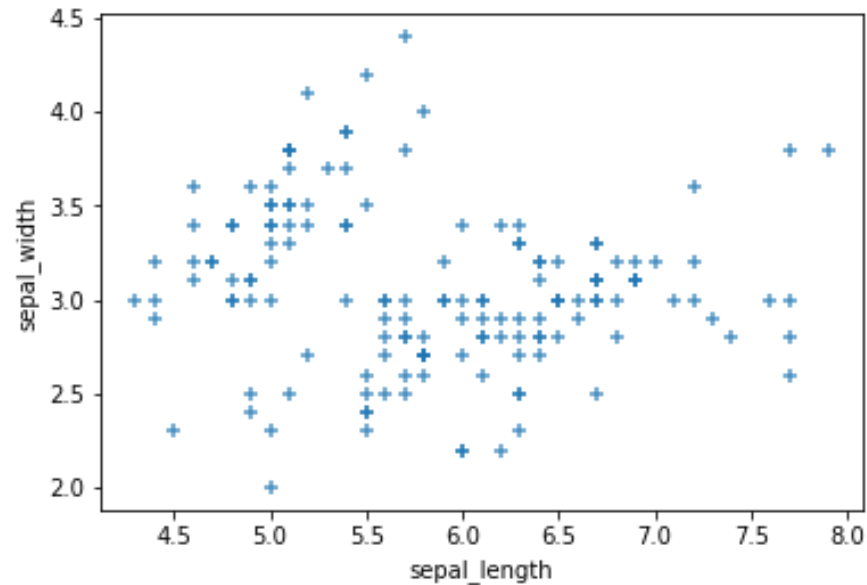


3. Maximize data density and the size of the data matrix, within reason

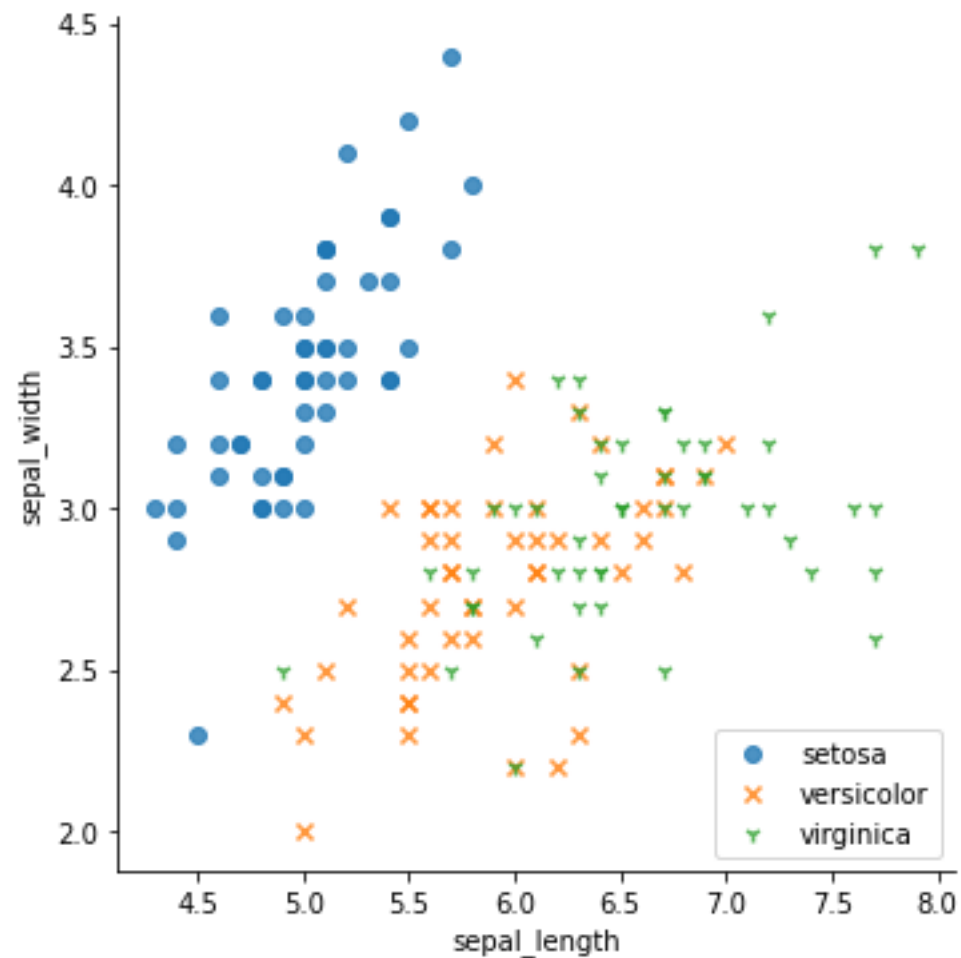
- High preformation graphics should be designed with special care. As the volume of data increases, data measures must shrink (smaller dots for scatters, thinner lines for busy time-series).



3. Maximize data density and the size of the data matrix, within reason

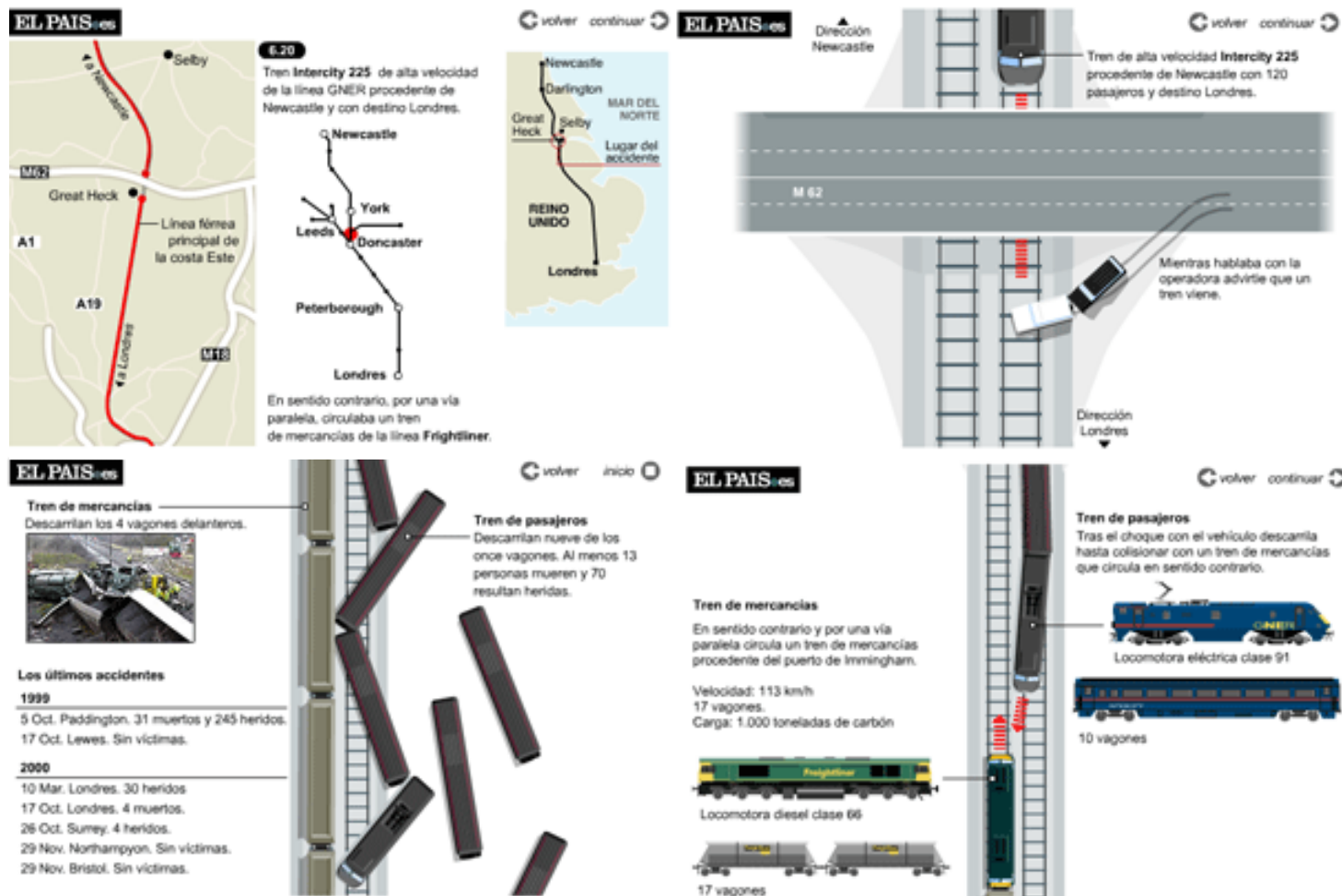


3. Maximize data density and the size of the data matrix, within reason



4. Establish Context

- Provide spatial, temporal and numerical context



Overview first, zoom and filter, details on demand

- influential mantra from Shneiderman

[\[The Eyes Have It: A Task by Data Type Taxonomy for Information Visualizations. Shneiderman. Proc. IEEE Visual Languages, pp. 336–343, 1996.\]](#)

- overview = summary
 - microcosm of full vis design problem

➔ Query

➔ Identify



➔ Compare



➔ Summarise

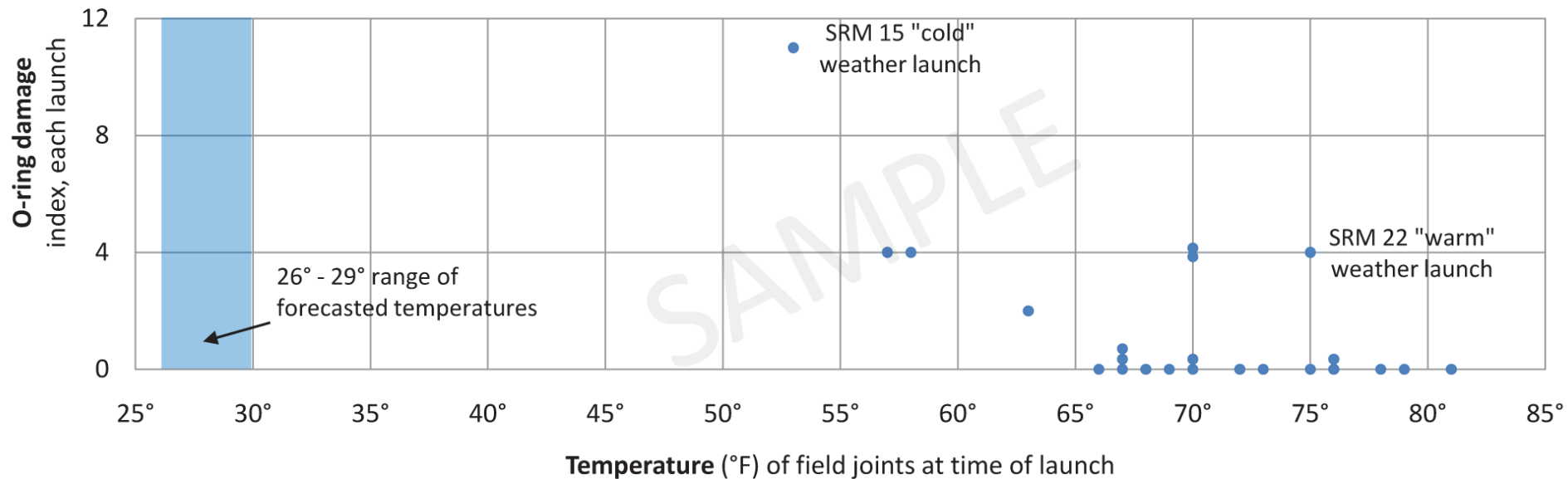


5. Show cause and effect, where possible

Space Shuttle History of Temperature and O-ring Damage

For All 24 Launches Prior to Challenger on January 28, 1986

Solid Rocket Motor (SRM) 15 and SRM 22 were the only prior launches discussed in relation to temperature on the eve of the launch.



Sources: Presidential Commission on the Space Shuttle Challenger Accident (PCSSCA) and Post-Challenger Evaluation of Space Shuttle Risk Assessment and Management as quoted in **Visual and Statistical Thinking** by Edward Tufte.

© Joe Bobcat

5. Show cause and effect, where possible



Magician Teller's definition of magic:

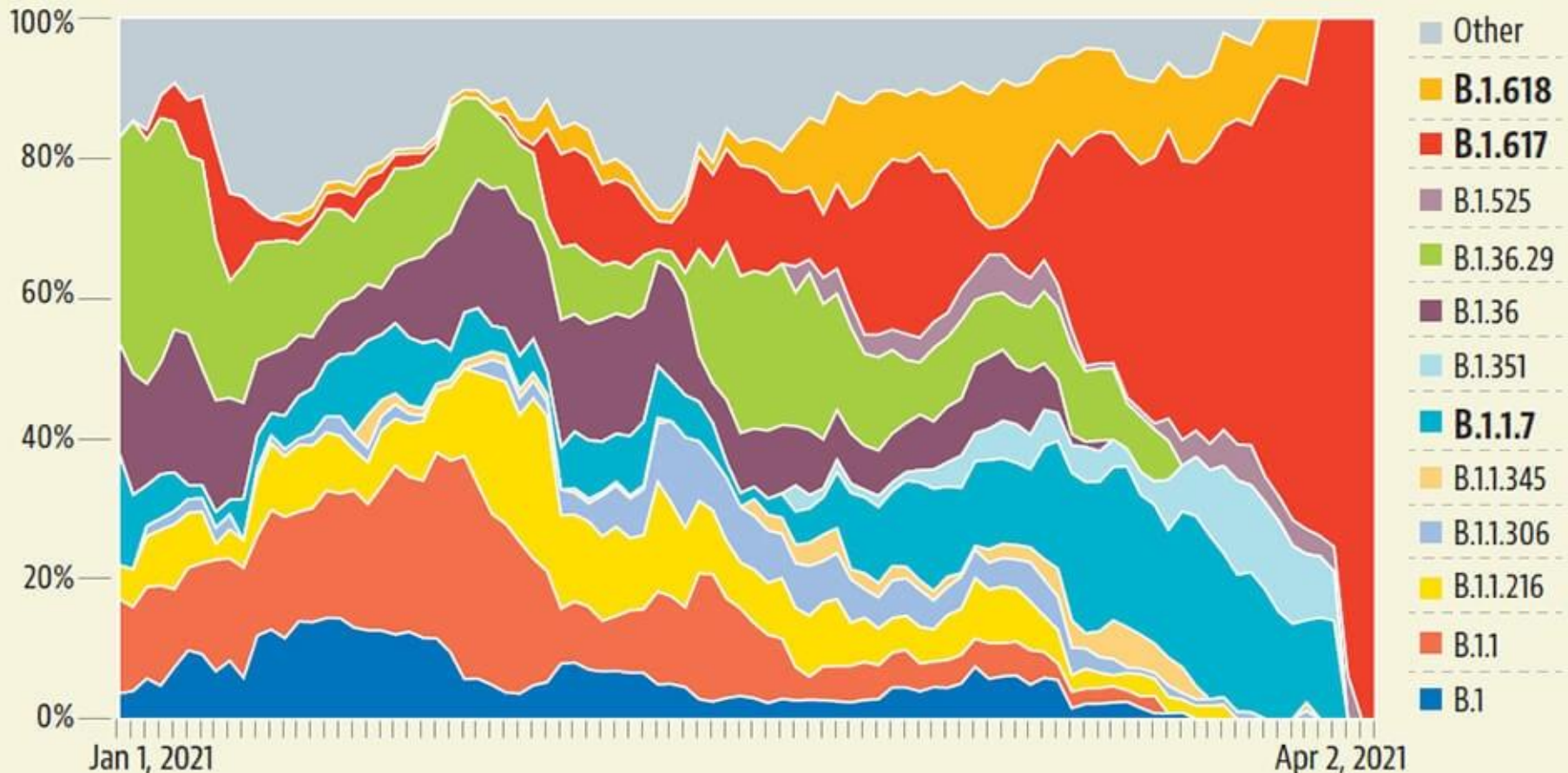
"The theatrical linking of a cause with an effect that has no basis in physical reality, but that — in our hearts — ought to."

6. Compare and contrast, utilize layering & separation

Second variant of interest?



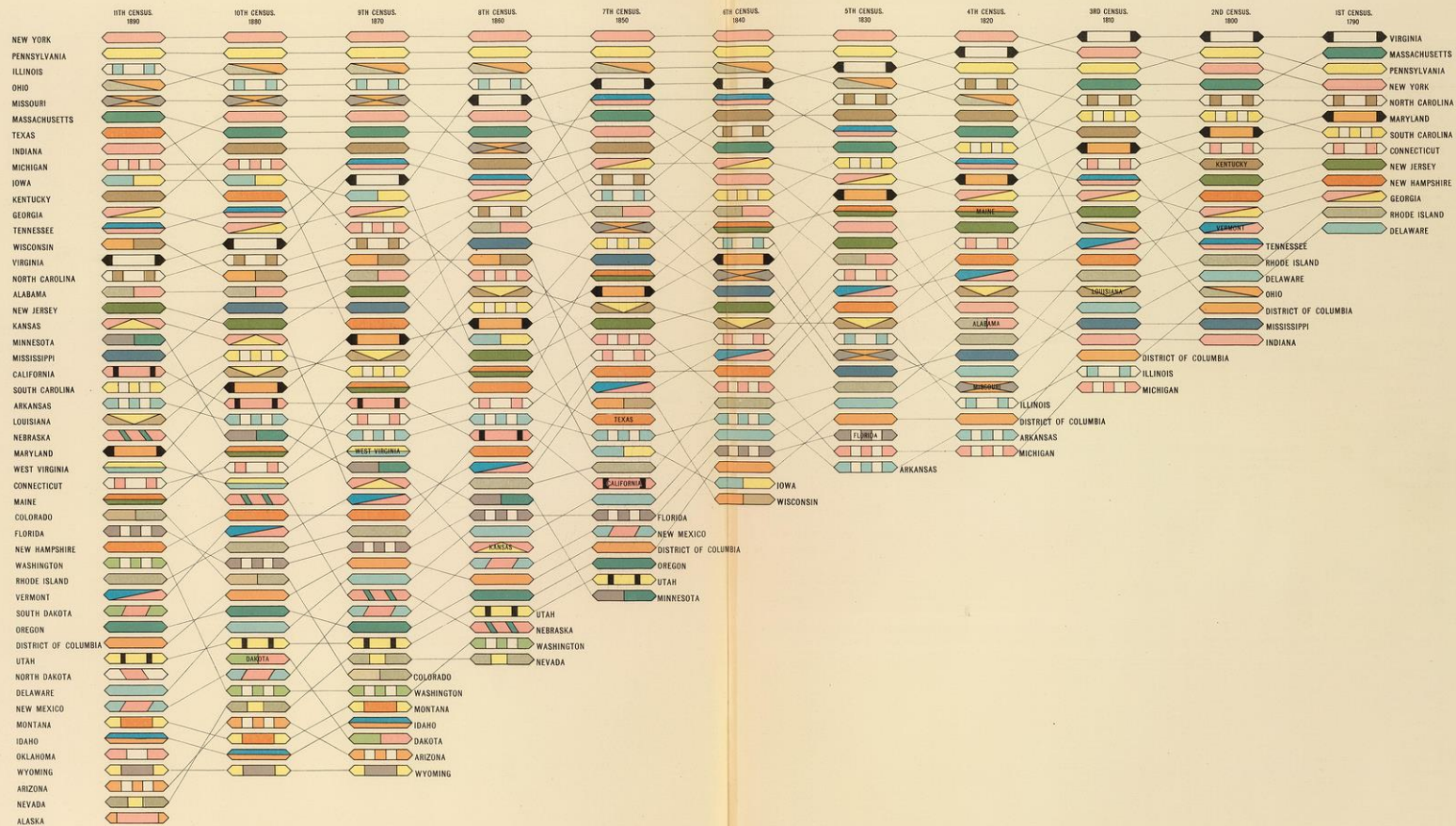
Sequences submitted from West Bengal show a large number of samples have the B.1.618 variant, although it appears to have plateaued and out-competed by B.1.617

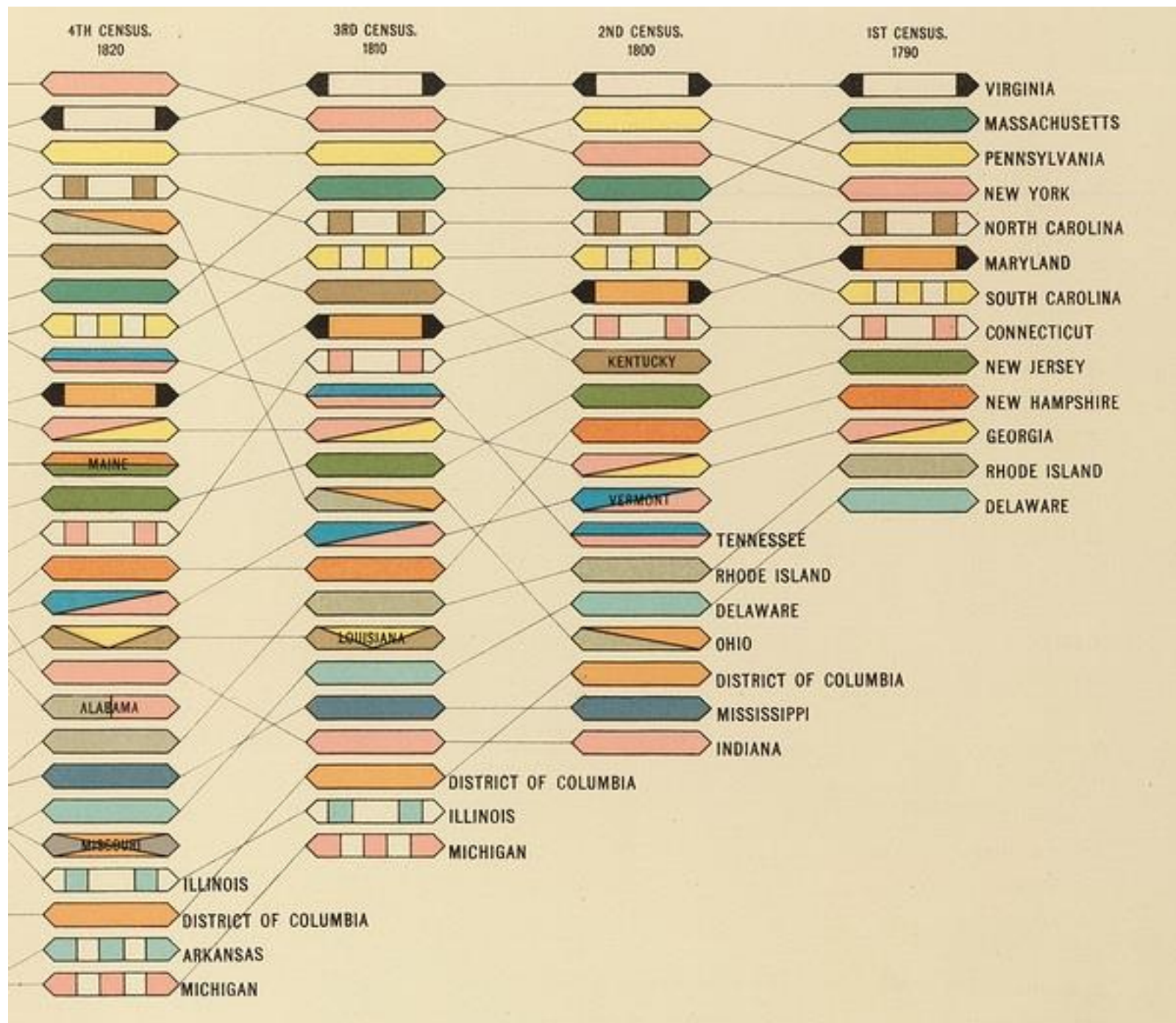


6. Compare and contrast, utilize layering & separation

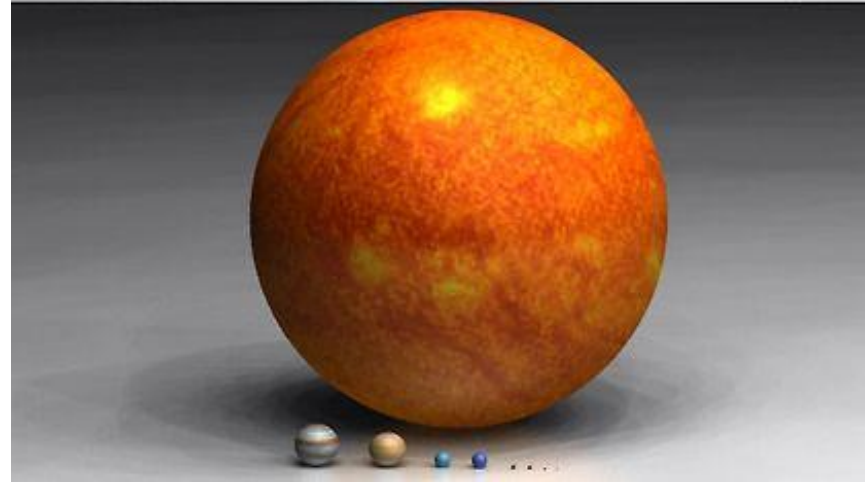
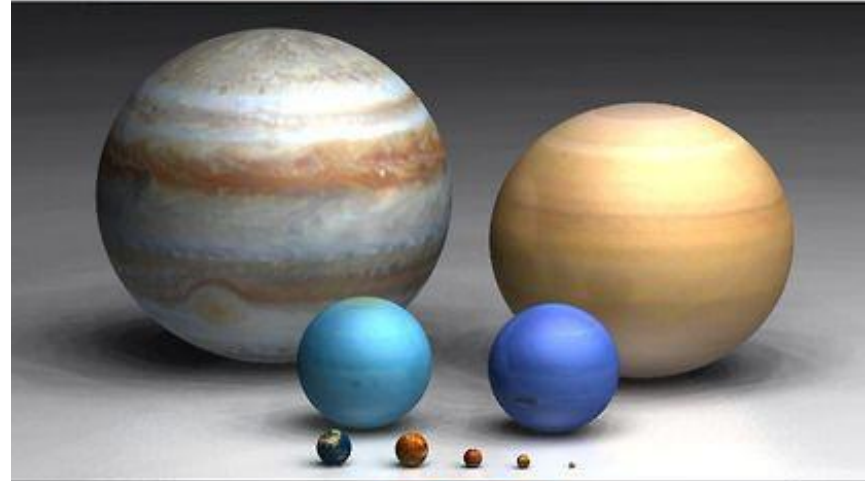
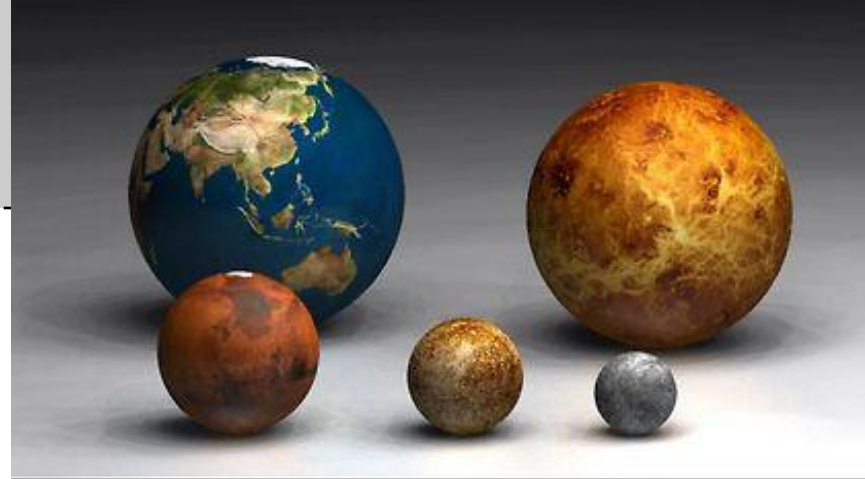
9. RANK OF STATES AND TERRITORIES IN POPULATION AT EACH CENSUS: 1790—1890.

PLATE 2.





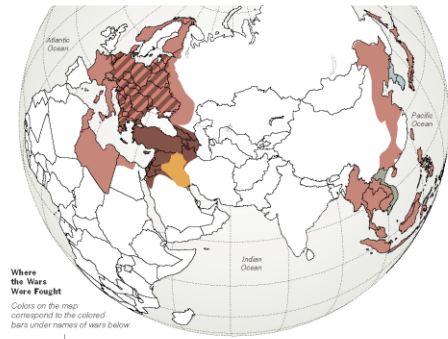
6. Compare and Contrast



6. Compare and Contrast

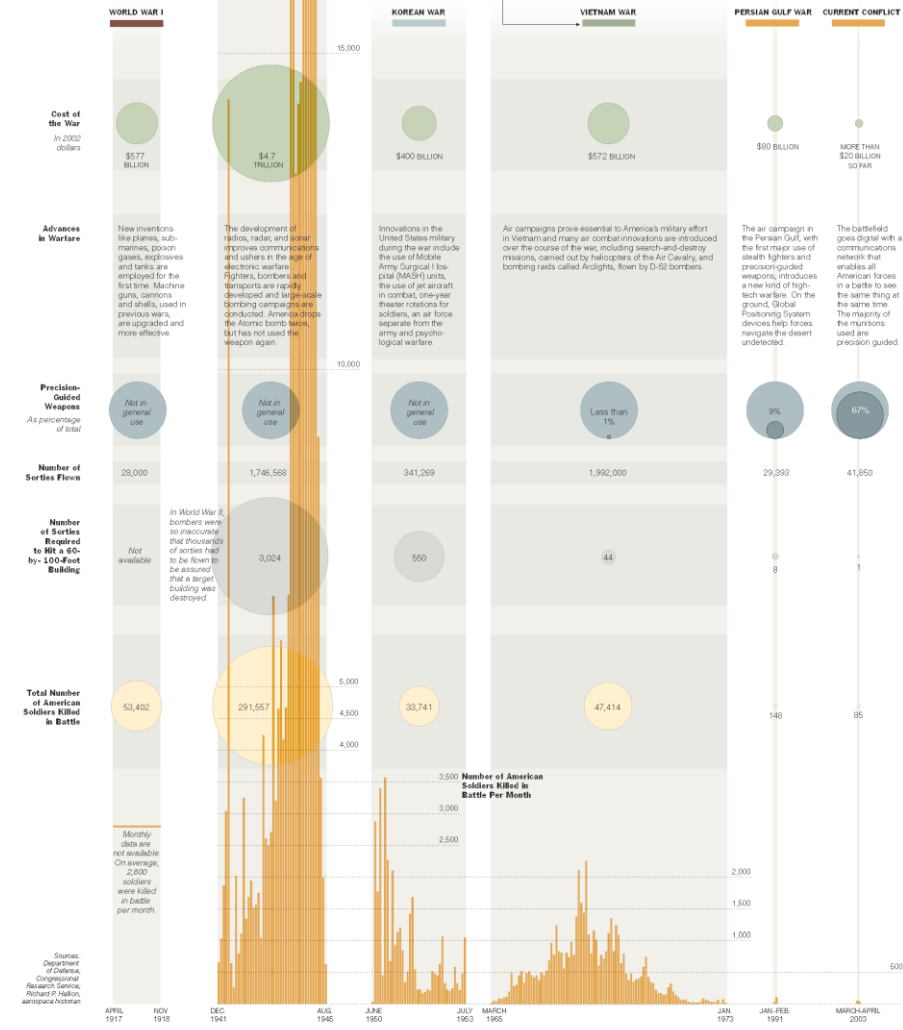
In Perspective: America's Conflicts

The current conflict in Iraq has ushered in the age of digital warfare. Over the years, as technological advances changed the way America fought its wars, casualties in the battlefield and the cost of war tended to fall.



Where the Wars Were Fought

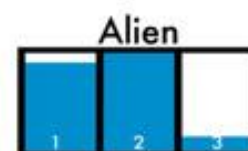
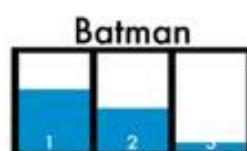
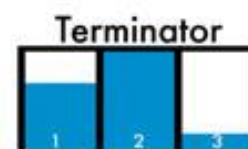
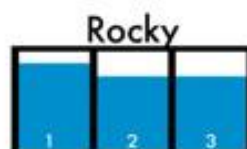
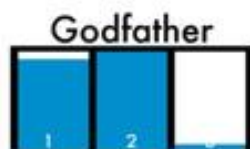
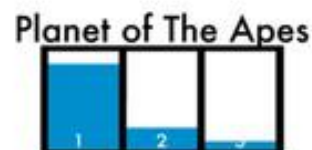
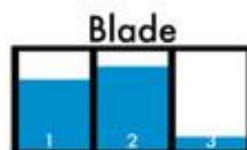
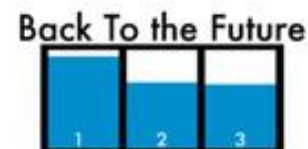
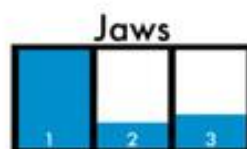
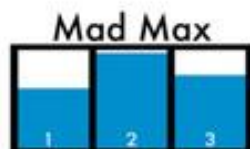
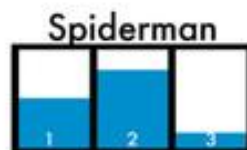
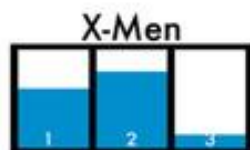
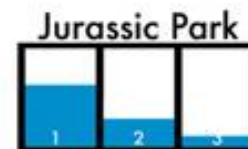
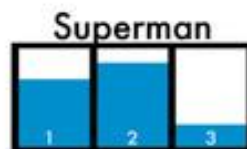
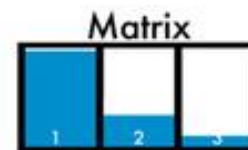
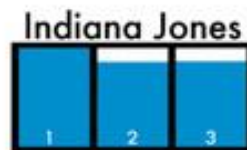
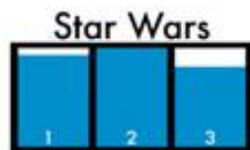
Colors on the map correspond to the colored bars under names of wars below



7. Escape flatland, use small multiples, parallel sequencing

- Data is multivariate doesn't necessarily mean 3D projection. How can we enhance multivariate data on inherently 2D surfaces? We can use small multiple graphs or parallel sequencing skill.

THE TRILOGY METER



2000: State-level support (orange) or opposition (green) on school vouchers, relative to the national average of 45% support



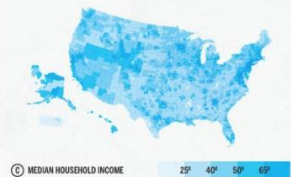
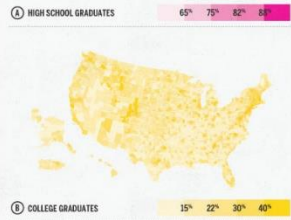
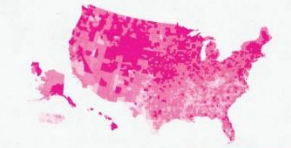
Orange and green colors correspond to states where support for vouchers was greater or less than the national average. The seven ethnoreligious categories are mutually exclusive. "Evangelicals" includes Mormons as well as born-again Protestants. Where a category represents less than 1% of the voters of a state, the state is left blank.



Separability vs. Integrality

READING, WRITING, AND EARNING MONEY

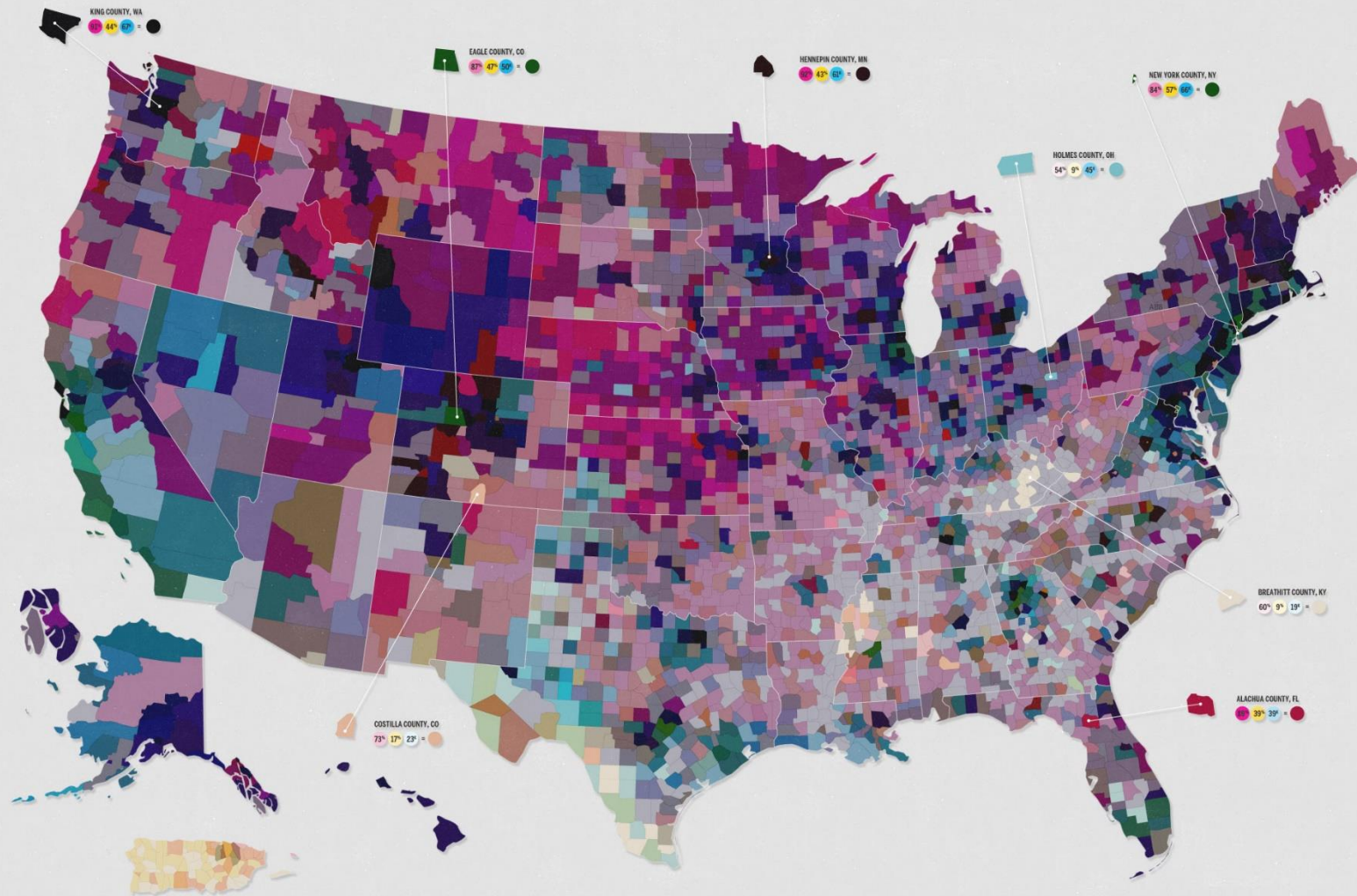
The latest data from the U.S. Census's American Community Survey paints a fascinating picture of the United States at the county level. We've looked at the educational achievement and the median income of the entire nation, to see where people are going to school, where they're earning money, and if there is any correlation.



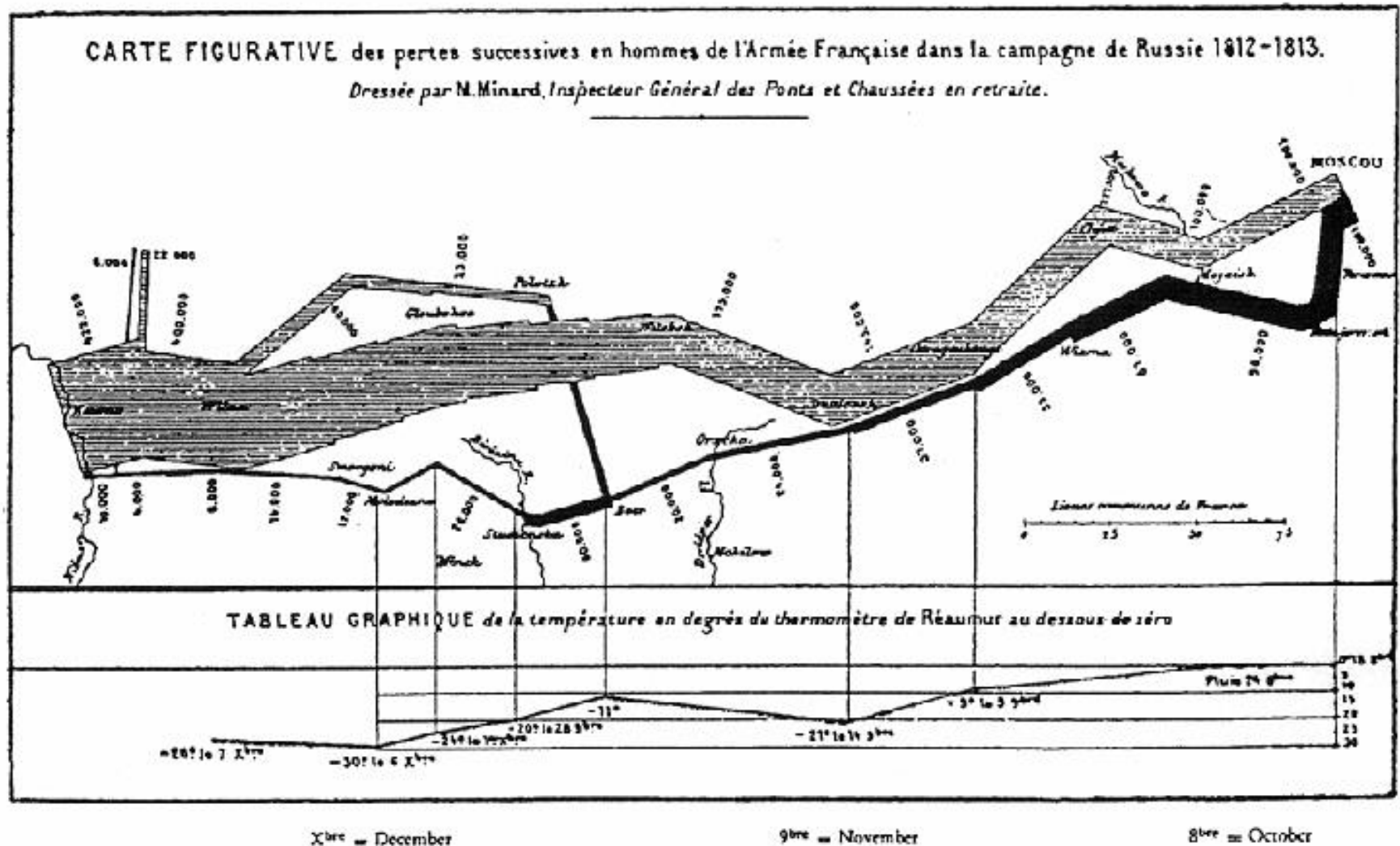
The map at right is a product of overlaying the three sets of data. The variation in hue and value has been produced from the data shown above. In general, darker counties represent a more educated, better paid population while lighter areas represent communities with fewer graduates and lower incomes.



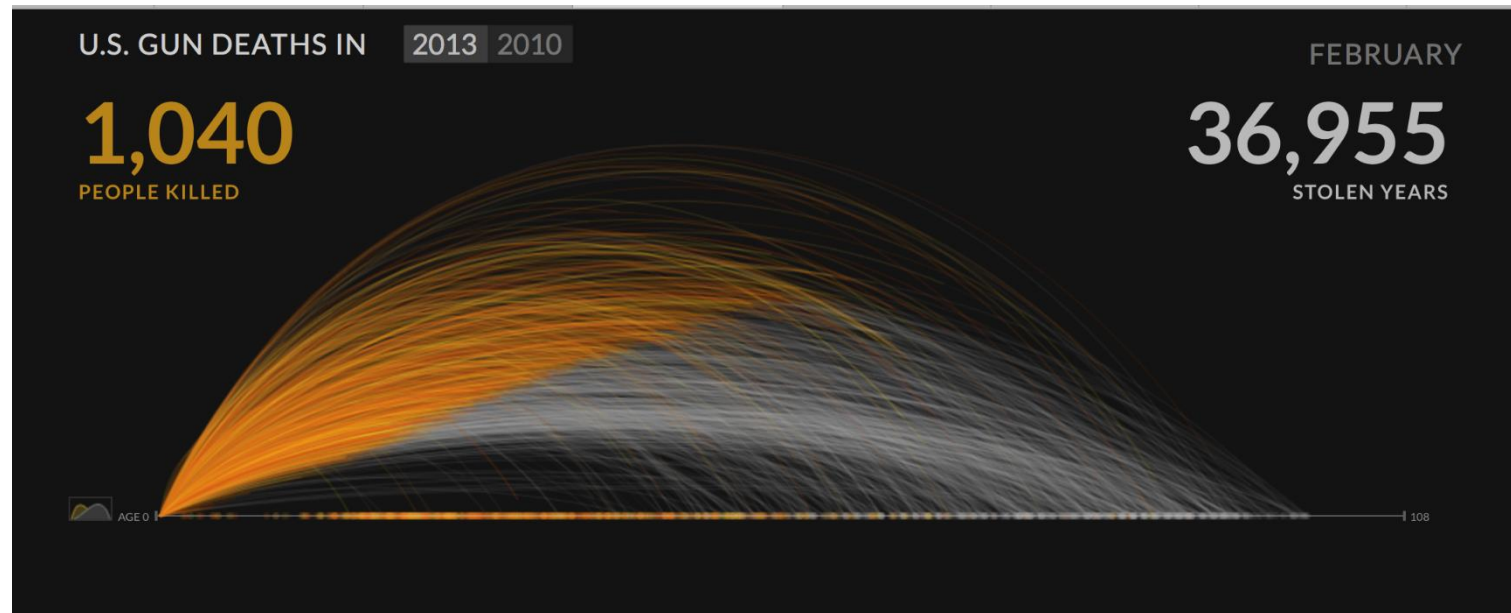
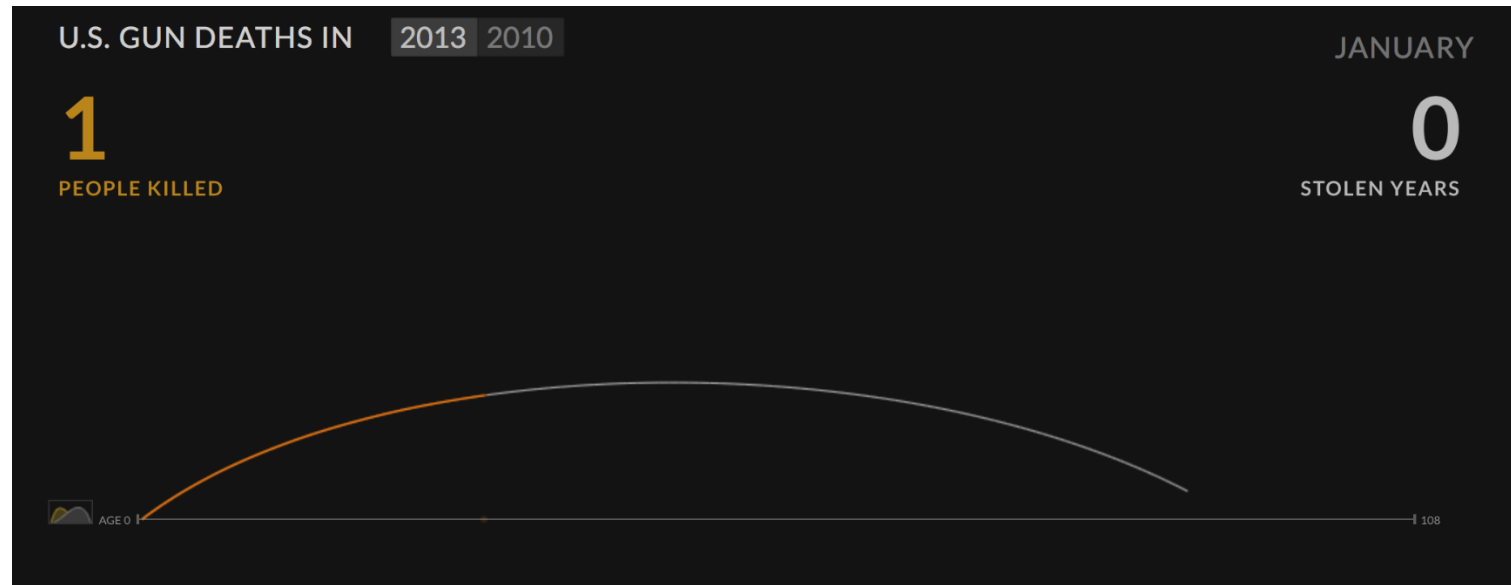
A collaboration between GOOD and Gregory Hebrack
SOURCE: US Census



8. Show Multiple Dimensions



9. Utilize narratives of space and time



9. Utilize narratives of space and time

Visualizing 500 Days of Summer

Created by [@rasagy](#) from NID Bangalore
More details about the process [on my blog!](#)
Share this on: [Twitter](#) [Facebook](#)

♡ 282

(Second IKEA visit)

*"Ah, hon? Our sink is broken.
Man, all of our sinks are broken."*

Day #0 Day #250 Day #500

Scene Length

● 10 sec

● 400 sec

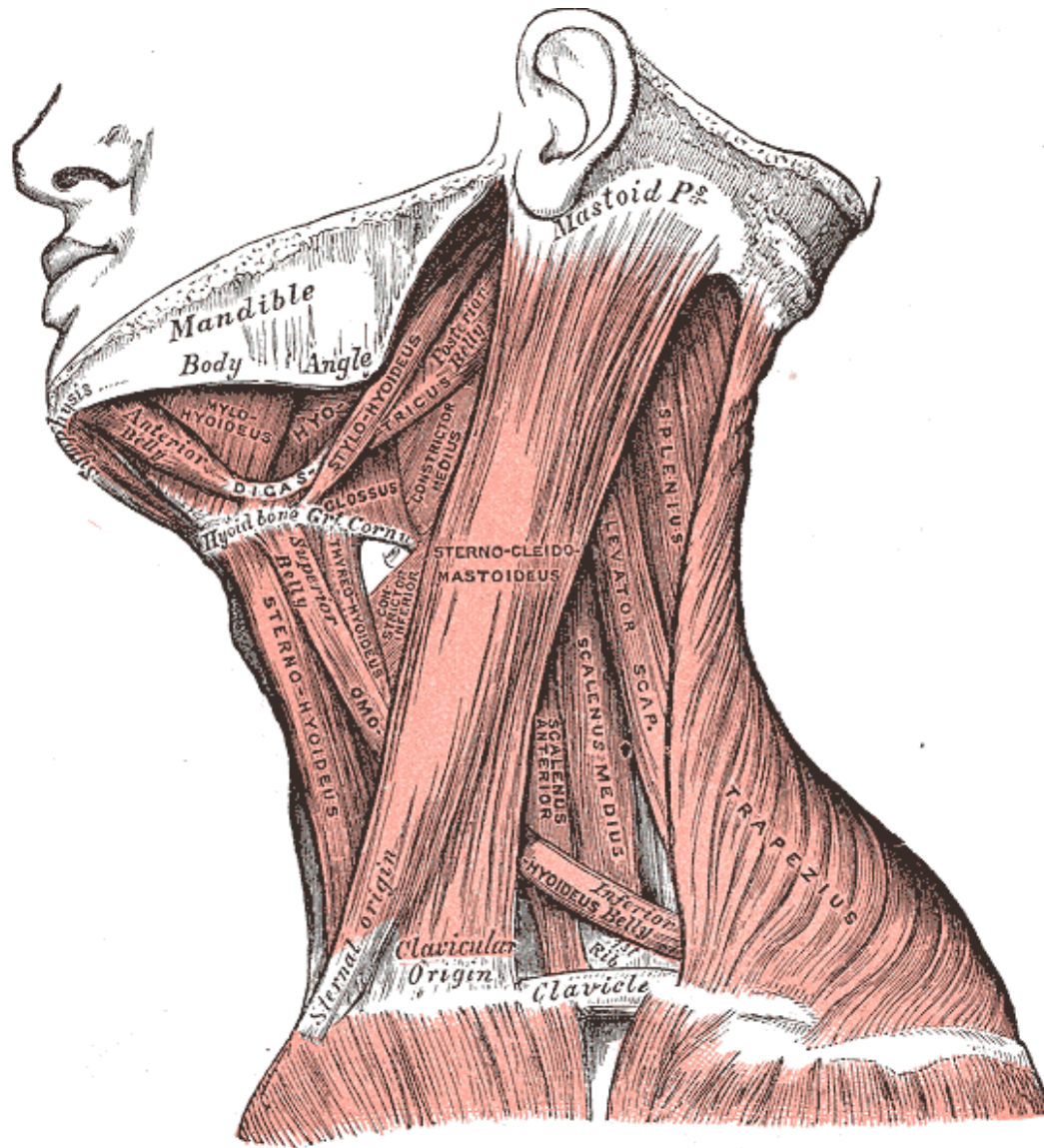
Relationship

♡ Together

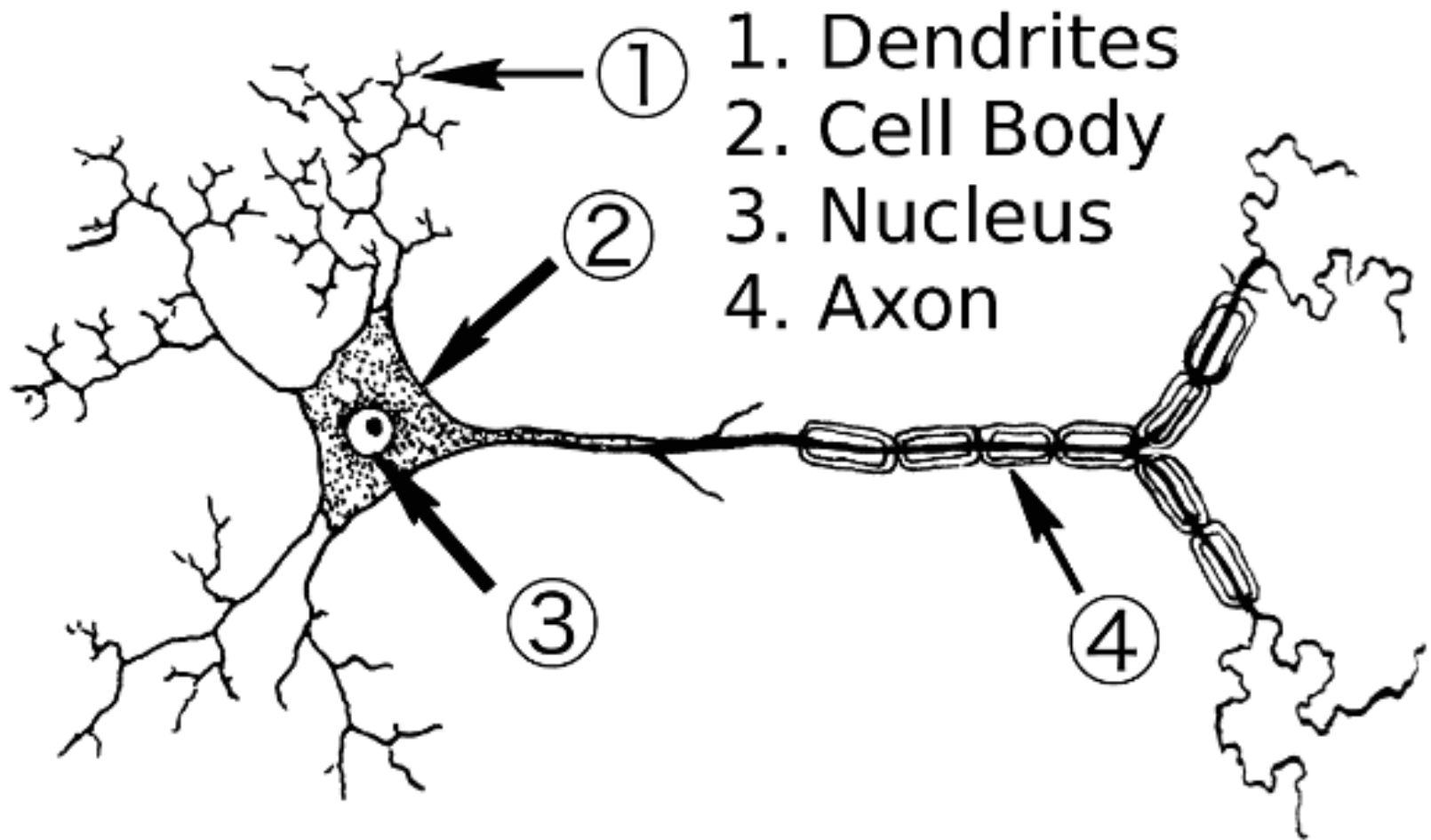
♡ Not together

♡ Acquaintance

10. Integrate image, number and text



10. Integrate image, number and text



Tufte's design principles for graphical excellence

1. Maximize the data-ink ratio, within reason
2. Mobilize every graphical element, perhaps several times over, to show the data
3. Maximize data density and the size of the data matrix, within reason
4. Establish context
5. Show cause and effect, where possible
6. Compare and contrast, utilize layering & separation
7. Escape flatland, use small multiples, parallel sequencing (reality is multivariate)
8. Show multiple dimensions
9. Utilize narratives of space and time
10. Integrate image, number and text